

Damping of breathing vortices in a dipole pair-plasma condensate: working notes

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The two box-scale vortices which breathe on the zonal condensate of a strongly driven, temperature-gradient-dominated dipole pair plasma do not persist: in a reduced-box simulation, they decay, ever faster, until they collapse. In these working notes, we assemble what can be said about why. We correct and complete an earlier calculation of the drift-resonant damping of a vortex on a frozen zonal state, show that it has a uniform-flow limit and a drift-free limit, in which the vortex obeys Rayleigh’s equation, and derive the drift-resonant damping at a critical layer. We then confront these calculations with the simulation, its mode-resolved energy budget and restarts of it. Three results emerge. Firstly, the vortices are the saturated shear-flow instability of their own jets: Rayleigh’s equation on the measured jets predicts their frequency and their radial structure, and their amplitude obeys the trapping law, viz., the trapping frequency is a constant multiple of the Rayleigh growth rate, while their energy falls by orders of magnitude. Secondly, their damping is linear and kinetic: electrostatic energy is converted into entropy at a rate that does not depend on amplitude, by Landau damping along the field line of the part of the vortices that is not flute-like, while the drift resonance is suppressed by trapping. Thirdly, the vortices decay because their jets slowly lose their instability, and they collapse when the jets become stable. We close by listing what remains open, in particular a first-principles calculation of the kinetic loss and of the erosion of the jets, and the simulations, now running, that bear on it.

1. Introduction

Strongly driven electron–positron plasmas in a dipole collapse, after a single explosive burst of interchange activity, into a box-scale zonal condensate on which two coherent vortices ride, one on each jet, carried in opposite directions and exchanging electrostatic energy with the zonal flow at twice their advection frequency — the ‘breathing’ of Kennedy & Plunk (2026*b*). Nothing in the collisionless transit-averaged model damps the flute-like zonal part of this state (Kennedy & Plunk 2026*a,b*), and the net injection of free energy by the gradients is negligible; the condensate is therefore long-lived, and the transport it carries is close to zero. The vortices, however, are not protected in the same way: in a reduced-box simulation of the same state, their energy falls slowly for several thousand time units, then ever faster, until they collapse into a weak, partly incoherent remnant. Coherent structures riding on sheared flows are common in zonally dominated plasma turbulence (Ivanov *et al.* 2020, 2025), and their lifetime can be as important as their existence. Since the vortices carry the large oscillating fluxes of the condensate, what removes them — and whether it would also remove them in a less idealised plasma — determines the long-time state of the system.

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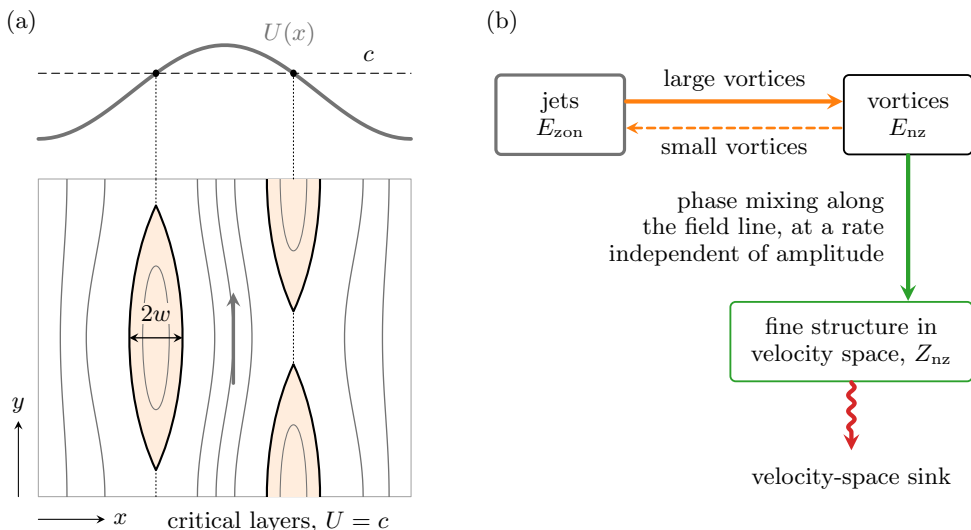


FIGURE 1. A cartoon of how the vortices lose their energy. (a) The geometry: the zonal flow $U(x)$ of one jet (grey, top) and the phase speed c of its vortex (dashed), which cross at two critical layers on the flanks of the jet (dotted); below, the streamlines in the frame of the vortex, with a cat's eye of trapped fluid, of half-width w , at each critical layer, the two rows of eyes staggered by half a wavelength, and the jet (grey arrow) meandering between them. (b) The flow of energy: while the vortices are large, the jets feed them (solid orange arrow), because the vortices are the shear instability of the jets; the electrostatic energy E_{nz} of the vortices is converted into entropy Z_{nz} , i.e., fine structure in velocity space, by phase mixing along the field line (green arrow), at a rate that does not depend on their amplitude, and removed there by the velocity-space sink (red arrow); once the jets have become stable, the exchange reverses (dashed orange arrow) and the vortices collapse.

The natural candidate is resonant phase mixing. The magnetic drift in a dipole is energy dependent, so a coherent structure moving through the plasma resonates with the particles whose drift (plus zonal advection) matches its speed, and loses energy to them in the manner of Landau damping (Landau 1946); the same resonance underlies the kinetic instabilities of the dipole pair plasma (Mishchenko *et al.* 2018; Kennedy *et al.* 2020). At finite amplitude, however, the vortex traps the resonant particles (or fluid elements), and since O'Neil (1965) it has been known that trapping suppresses resonant damping once the trapping frequency exceeds the linear damping rate. A nonlinear decay whose rate increases as the amplitude falls is what this picture suggests, and it was the hypothesis of an earlier analytical note (unpublished, dated 30 September 2026; hereafter ‘the earlier note’), on which the present notes build. That note left four questions open, viz., (i) whether the resonance is really with the drift; (ii) whether the damping has the sign of a damping; (iii) whether the vortices sit where the note assumed; and (iv) whether the observed acceleration is the trapping crossover or something else. We shall find that it is something else.

The picture that emerges is sketched in figure 1. Each vortex has its critical layers on the flanks of its jet, with a cat's eye of trapped fluid at each of them. The vortices are the saturated shear-flow instability of the jets, so the jets feed them; they lose their energy, at a rate that does not depend on their amplitude, to fine structure in velocity space, which is then dissipated; and they decay as the jets slowly lose their instability, until the jets become stable, the exchange reverses, and the vortices collapse.

In these notes, we first summarise the setting in §2, viz., the transit-averaged equations,

the strongly driven condensate and the reduced-box simulation with its restarts, and then, in §3, the observations that are to be explained. In §4, we derive the linear response of a vortex on a frozen zonal state, correcting the treatment of the zonal distribution, and show that it has two limits: a uniform-flow limit, which we work out in closed form, and a drift-free limit, in which the vortex obeys Rayleigh’s equation and the resonance is a hydrodynamic critical layer; we also derive the effect of a weak drift at such a layer. In §5, we turn to trapping, both on the flanks of a jet and at its maximum, and to the interplay of phase mixing with the hyperdiffusive sinks of the simulation. In §6, we confront these calculations with the simulation data, before finally stating, in §7, what is established and what is open. Appendix A collects the energy integrals of the uniform-flow limit.

2. The setting

In this section, we recall the equations of Kennedy & Plunk (2026b) and Kennedy & Plunk (2026a) to the extent needed here, describe the strongly driven condensate in which the vortices live, and specify the simulations analysed in §6.

2.1. Transit-averaged equations

We consider a symmetric electron–positron plasma, with charges $e_p = -e_e = e$, a common Maxwellian f_0 of density n_0 and temperature T , in the ordering $\rho \ll \lambda_D \ll L$, where ρ is the Larmor radius, $\lambda_D = \sqrt{\epsilon_0 T / 2n_0 e^2}$ the Debye length and L an equilibrium length. We use flux coordinates (ψ, α, l) with $\mathbf{B} = \nabla\psi \times \nabla\alpha$, and write $\delta f_a = h_a - e_a \phi f_0 / T$. Below the transit and bounce frequencies, $h_a(\psi, \alpha, \varepsilon, \lambda, t)$ is independent of l (Helander 2014; Mishchenko *et al.* 2018), where $\varepsilon = mv^2/2$ is the energy and $\lambda = v_\perp^2 / Bv^2$ the pitch-angle variable, and particles respond to the orbit-averaged potential

$$\bar{\phi}(\lambda) = \frac{1}{\hat{\tau}(\lambda)} \int_{l_1}^{l_2} \frac{\phi(l) dl}{\sqrt{1 - \lambda B(l)}}, \quad \hat{\tau}(\lambda) = \int_{l_1}^{l_2} \frac{dl}{\sqrt{1 - \lambda B(l)}}, \quad (2.1)$$

where l_1 and l_2 are the bounce points of a trapped particle or the ends of one field-line period for a passing one. To leading order in $k_\perp \rho$, the kinetic and field equations are (Abel *et al.* 2013; Mishchenko *et al.* 2018; Kennedy & Plunk 2026a)

$$\partial_t h_a + \{\bar{\phi}, h_a\} + \bar{v}_{da} \partial_\alpha h_a = \frac{e_a f_0}{T} (\partial_t \bar{\phi} - v_{*a}^T \partial_\alpha \bar{\phi}), \quad (2.2)$$

$$(1 - \lambda_D^2 \nabla_\perp^2) \frac{e\phi}{T} = \frac{1}{2n_0} \int d^3\mathbf{v} (h_p - h_e), \quad (2.3)$$

where $\{X, Y\} \equiv \partial_\psi X \partial_\alpha Y - \partial_\alpha X \partial_\psi Y$ is the $\mathbf{E} \times \mathbf{B}$ advection, $\bar{v}_{da}$ is the orbit-averaged binormal magnetic drift, which is energy dependent and of opposite sign for the two species, and $v_{*a}^T = -(T/e_a) \partial_\psi \ln f_0$ is the gradient drive at fixed energy. The dipole has no radial magnetic drift, so the zonal flow suffers no collisionless damping.

The nonlinearity of (2.2) conserves the free energy W and the electrostatic energy E separately, and their difference $Z = W - E \geq 0$ is the entropy of the orbit-averaged perturbation (Kennedy & Plunk 2026a,b); these invariants are the closed-field-line counterparts of those of two-dimensional gyrokinetics (Plunk *et al.* 2010; Plunk & Tatsuno 2011). Here, we need only two further facts. Firstly, the free energy obeys

$$\frac{dW}{dt} = P - D, \quad (2.4)$$

where P is the injection by the gradients and $D \geq 0$ is the irreversible removal, which, in the simulations, is entirely due to hyperdiffusion (Krommes & Hu 1994; Sugama *et al.*

1996). Secondly, if we split $E = E_{\text{zon}} + E_{\text{nz}}$ into its zonal ($k_y = 0$) and non-zonal parts, the nonlinear transfer between the two satisfies

$$(\partial_t E_{\text{zon}})_{\text{NL}} = -(\partial_t E_{\text{nz}})_{\text{NL}}, \quad (2.5)$$

i.e., the zonal flow and the vortices exchange electrostatic energy without loss; this exchange is the breathing (Kennedy & Plunk 2026b). We shall write Z_{nz} and $W_{\text{nz}} = E_{\text{nz}} + Z_{\text{nz}}$ for the non-zonal parts of Z and W , and $\Gamma_E \equiv -d \ln E_{\text{nz}}/dt$ for the decay rate of the non-zonal electrostatic energy. For a structure of fixed shape, $\Gamma_E = 2\gamma_{\text{amp}}$, where γ_{amp} is the decay rate of its amplitude.

2.2. The strongly driven condensate

At $\eta \equiv L_n/L_T = 1$, the drive is above the fluid-interchange threshold (Mishchenko *et al.* 2018), and the interchange grows at the same rate at every wavelength the box allows. A single $k_y = 1$ streamer then collapses, in one burst, into $k_x = 1$ zonal bands, viz., one jet in each direction across the periodic box, modulated by two box-scale vortices, one on each jet (Kennedy & Plunk 2026b). The two vortices move in opposite directions, related by the symmetry of the pair-plasma equations under exchange of the species together with $\phi \rightarrow -\phi$ and $y \rightarrow -y$, and their relative phase sets the tilt of the flow between them, so the exchange (2.5) runs at $\omega_b = 2|\omega_{\text{adv}}|$, twice the advection frequency of either vortex. Over a breathing cycle, the state changes little; over many cycles, the vortices decay. In what follows, we call $k_y = 1$ the vortex and $k_y = 0$ the jet, and the question of these notes is why the former decays while the latter does not.

2.3. The reduced-box simulation and its restarts

All the data in these notes comes from one nonlinear simulation with the gyrokinetic code GENE (Jenko *et al.* 2000) in the dipole flux tube of Kennedy & Plunk (2026b), at $\eta = 1$, started from noise, in a box a quarter of the production box in each perpendicular direction, together with restarts of it. Its parameters are collected in table 1. There is no collision operator: all irreversible removal is by fourth-order hyperdiffusion, spectral in x and y [rate $\nu_{\perp}(\Delta x k_x/2)^4$, where Δx is the radial grid spacing, and likewise in y], and by finite-difference hyperdiffusion along the field line and in $v_{\parallel}$. One property of the grid will matter in §6.5: the radial grid spacing is comparable with the Debye length, $\lambda_D \approx 1.2 \Delta x$. The run is continued for many thousands of breathing periods after the burst, until long after the vortices have disappeared. Times are in $L_{\text{ref}}/c_{\text{ref}}$, lengths in ρ_{ref} , and energies in the units of the GENE free-energy diagnostics (Bañón Navarro *et al.* 2011, 2016).

The first two restarts, which we call the *amplitude tests*, start from a checkpoint of the original run in the slow phase of the decay, with every $k_y \geq 1$ component of the distribution function multiplied by $\epsilon = 0.3$ or $\epsilon = 0.03$ and the zonal component left untouched; the potential follows from (2.3). Each is run for a few hundred breathing periods. If the vortex decay were linear on a frozen zonal state, these restarts would decay at exactly the same rate as the original, since the normalised linear evolution of a rescaled perturbation is unchanged; this is the null hypothesis that the tests were designed to probe.

Five further restarts start from a later checkpoint, in the accelerating phase of the decay, and change one thing each: the velocity-space hyperdiffusion or the perpendicular hyperdiffusion, multiplied or divided by four, or the radial resolution, doubled at fixed box size and fixed hyperdiffusion coefficient. Since the perpendicular hyperdiffusion is normalised to the grid, the last of these also weakens the radial hyperdiffusion at a given

Quantity	Symbol	Value
Gradient drives	$\omega_n = \omega_T$	2.7816 ($\eta = 1$)
Grid	$n_{x0} \times n_{ky0} \times n_{z0}$	$64 \times 16 \times 32$
Velocity grid	n_v, L_v	84, 4.5
Box	$L_x, k_{y,\min}$	$3689 \rho_{\text{ref}}, 2 \times 10^{-3} \rho_{\text{ref}}^{-1}$
Debye length	λ_D	$70.7 \rho_{\text{ref}} = 1.23 \Delta x$
Perpendicular hyperdiffusion	$\nu_x = \nu_y$	2.4×10^{-3} (spectral, fourth order)
Parallel hyperdiffusion	ν_z	0.05 (fourth order)
Velocity-space hyperdiffusion	ν_v	0.2 (fourth order in $v_{\parallel}$)
Collisions	—	none

TABLE 1. A summary of the parameters of the reduced-box simulation analysed in these notes. The restarts share them, except where stated otherwise.

physical wavenumber, by a factor of sixteen: it is a test of the grid scale against the Debye scale, not a convergence test. These restarts are running at the time of writing, and only the questions that they are to answer appear in these notes.

3. Observations

Before turning to the theory, let us state what is to be explained, leaving the detailed measurements to §6.

The energy E_{nz} of the vortices falls by orders of magnitude between the burst and its collapse [see figure 2(a)]. The decay is not exponential. After a relaxation that follows the burst, the rate Γ_E passes through a broad minimum and then increases, ever faster, until the vortices collapse [see figure 2(b)]; there is no plateau. After the collapse, a weak remnant persists, decaying slowly and irregularly. Throughout, the zonal energy E_{zon} decays at a rate that is smaller than Γ_E by orders of magnitude, and the net injection P is negligible.

The amplitude tests decay faster than the original, not at the same rate, and the smaller the amplitude, the faster the decay [see figure 3(a)]. Both restarts first grow, relative to their rescaled initial energy, before decaying.

4. Linear theory of the vortex on a frozen zonal state

In this section, we derive the linear response of a small vortex on a frozen zonal state, correct the treatment of the zonal distribution in the earlier note, and show that the resulting problem has two limits that behave entirely differently. The algebra of §4.1 and §4.2 was rederived independently and checked numerically; the reductions of §4.4 are introduced in these notes.

4.1. Linear response

Assuming that the zonal state evolves slowly compared with the vortex, we freeze a flute-like zonal state (Φ, H_a) and consider a small travelling perturbation,

$$\phi = \Phi(\psi) + \varphi(\psi, l) e^{ik\alpha - i\omega t}, \quad h_a = H_a(\psi, \varepsilon, \lambda) + g_a e^{ik\alpha - i\omega t}, \quad (4.1)$$

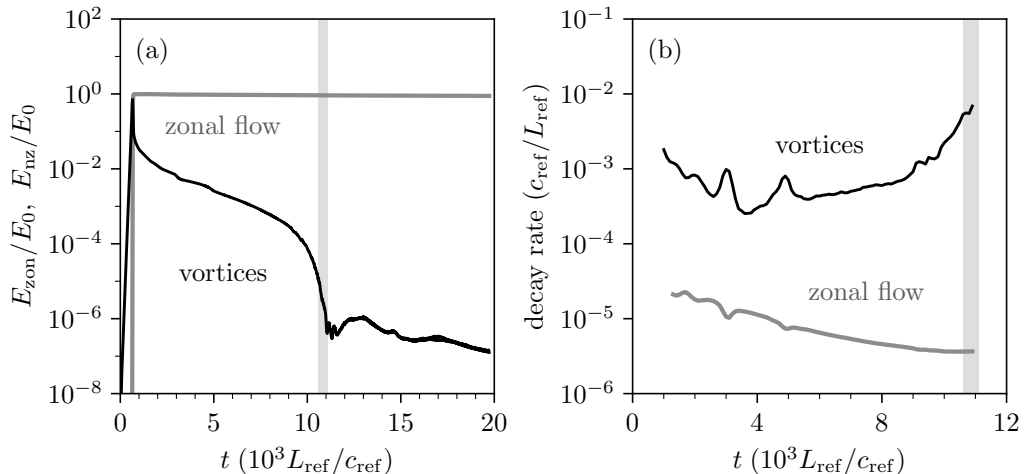


FIGURE 2. Electrostatic energy of the zonal flow and of the vortices in the reduced-box simulation. (a) The zonal energy E_{zon} (grey) and the non-zonal energy E_{nz} (black), as running means over about ten breathing periods, in units of E_0 , the largest value of E_{zon} . (b) The decay rates $\Gamma_E = -d \ln E_{\text{nz}}/dt$ (black) and $-d \ln E_{\text{zon}}/dt$ (grey), from least-squares fits of the logarithm of each energy in sliding windows of about a hundred breathing periods, from the end of the relaxation that follows the burst to the end of the collapse. The grey band marks the collapse in this and all subsequent figures. The vortices lose many decades of energy, ever faster and without a plateau, and then collapse, while the zonal flow barely changes.

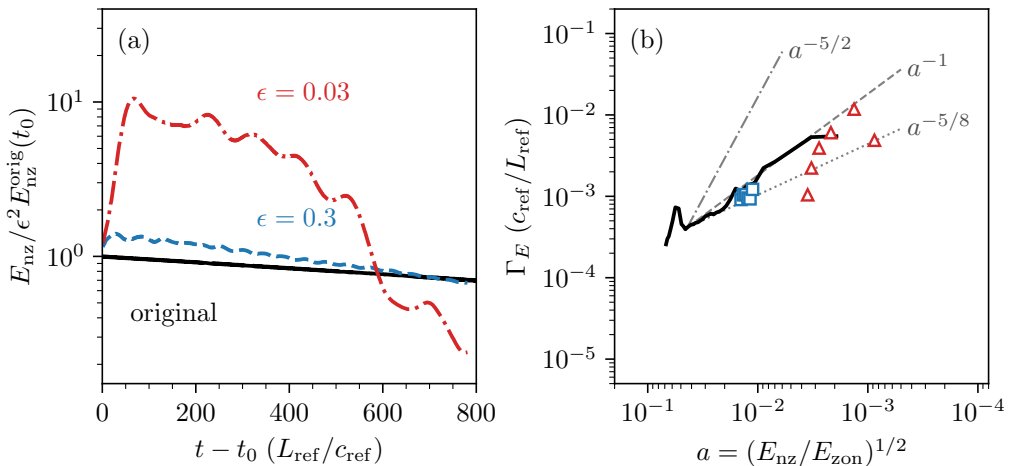


FIGURE 3. Amplitude dependence of the vortex decay. (a) The non-zonal electrostatic energy of the original run (black solid) and of the restarts with $\epsilon = 0.3$ (blue dashed) and $\epsilon = 0.03$ (red dash-dotted), divided by ϵ^2 and by the energy of the original run at the restart time t_0 , as running means over about ten breathing periods. (b) The decay rate Γ_E against the amplitude measure $a = (E_{\text{nz}}/E_{\text{zon}})^{1/2}$, which decreases to the right: the original run (black line; fits in windows of about a hundred breathing periods, from the end of the relaxation that follows the burst to the collapse) and the restarts after their initial transient (blue squares and red triangles; windows half as long). The grey lines are the power laws $a^{-5/8}$ (dotted), a^{-1} (dashed) and $a^{-5/2}$ (dash-dotted) of §5.4, all drawn through the original run at the restart time. A linear process would keep the three curves of (a) together; instead, the weaker the vortices, the faster they decay, at a rate that follows roughly one curve for all three runs, between $a^{-5/8}$ and a^{-1} and far from $a^{-5/2}$.

where real parts are understood, k is the binormal wavenumber, and the perturbation need not be flute-like. Let $U \equiv \partial_\psi \Phi$ be the zonal advection velocity in flux coordinates.[†] The pair (Φ, H_a) is an exact steady solution of (2.2), since $\{\Phi, H_a\} = 0$ for flute-like functions of ψ alone. With $\{\Phi, g_a e^{ik\alpha}\} = ikU g_a e^{ik\alpha}$ and $\{\bar{\varphi} e^{ik\alpha}, H_a\} = -ikH'_a \bar{\varphi} e^{ik\alpha}$, where the prime denotes ∂_ψ at fixed (ε, λ) , linearisation of (2.2) gives the particle response

$$g_a = \frac{\mathcal{N}_a(\omega)}{\omega - \Omega_a} \bar{\varphi}, \quad \Omega_a = k(U + \bar{v}_{da}), \quad \mathcal{N}_a(\omega) = \frac{e_a f_0}{T} \omega - k \partial_\psi (f_0 + H_a), \quad (4.2)$$

where we have used the identity $(e_a f_0 / T) v_{*a}^T = -\partial_\psi f_0$, so that the drive is the total radial gradient of the l -independent distribution $f_0 + H_a$.

The earlier note kept H'_a but did not identify its content, and that content matters. The zonal part of (2.3) forces H_a to contain a Boltzmann part,

$$H_a = \frac{e_a f_0}{T} \Phi + \delta F_a, \quad (4.3)$$

where $\delta F_a(\psi, \varepsilon, \lambda)$ is the zonal modification of the distribution, i.e., the flattening (or steepening) of the gradients by the condensate. Then, to leading order in the ratio of the zonal scale to the equilibrium scale,[‡]

$$\mathcal{N}_a(\omega) = \frac{e_a f_0}{T} [\omega - kU + k \tilde{v}_{*a}(\varepsilon, \lambda, \psi)], \quad \frac{e_a f_0}{T} \tilde{v}_{*a} \equiv -\partial_\psi (f_0 + \delta F_a). \quad (4.4)$$

In other words, the drive depends on ω only through the Doppler-shifted frequency $\tilde{\omega} \equiv \omega - kU$: a locally uniform zonal flow, $\Phi = U\psi$, is a pure Doppler shift, as it must be, and the effective diamagnetic drift $\tilde{v}_{*a}$ is built from the gradients of the distribution as modified by the condensate. Without the Boltzmann part of (4.3), the response (4.2) would not be Doppler invariant. Note also that the zonal part of (2.3) constrains the admissible frozen states: when $|\nabla\psi|^2$ varies along the field line, a flute-like Φ requires the charge moment of $\delta F_p - \delta F_e$ to vary with l accordingly.

4.2. The field equation and the resonant contribution

Let us define $\mathcal{P}_k \equiv 1 - \lambda_D^2 \nabla_{\perp, k}^2$, with ∂_α replaced by ik . Substituting (4.2) into (2.3) gives the mode equation

$$\mathcal{L}(\omega) \varphi \equiv \mathcal{P}_k \varphi - \frac{T}{2n_0 e^2} \sum_a e_a \int d^3 \mathbf{v} \frac{\mathcal{N}_a(\omega)}{\omega - \Omega_a} \bar{\varphi} = 0, \quad (4.5)$$

where the velocity integral is evaluated at the position along the field line at which (2.3) is imposed and the potential inside it is orbit averaged, so that the coupling between l and λ is retained. For the convention $e^{-i\omega t}$, causality requires starting from $\text{Im } \omega > 0$ and continuing analytically; at real frequency, this gives the Landau prescription (Landau 1946)

$$\frac{1}{\omega_r - \Omega + i0} = \text{PV} \frac{1}{\omega_r - \Omega} - i\pi \delta(\omega_r - \Omega). \quad (4.6)$$

[†] Not to be confused with the specific flux-tube volume $U = \oint dl/B$ of Kennedy & Plunk (2026b), which does not appear in these notes.

[‡] Strictly speaking, $\partial_\psi H_a$ also contains $(e_a \Phi / T) \partial_\psi f_0$. Compared with $(e_a f_0 / T) \partial_\psi \Phi$, this is smaller by the ratio of the radial scale of the zonal flow to the equilibrium gradient length, which is small in a local flux tube; we neglect it throughout.

Accordingly, we split the operator as $\mathcal{L} = \mathcal{L}_R + i\mathcal{L}_I$, where $\mathcal{L}_R$ is the principal-value part and $\mathcal{L}_I$ the delta-function part,

$$\mathcal{L}_I(\omega_r)\varphi = \frac{\pi T}{2n_0 e^2} \sum_a e_a \int d^3\mathbf{v} \mathcal{N}_a(\omega_r) \bar{\varphi} \delta(\omega_r - k[U + \bar{v}_{da}]). \quad (4.7)$$

Note that this is a split of the operator, not of $\mathcal{L}\varphi$ into real and imaginary parts, since φ is complex. The delta function selects the drift-resonant particles, $\omega_r/k = U(\psi) + \bar{v}_{da}(\varepsilon, \lambda)$; this is the same resonance that drives the kinetic instabilities of the dipole (Mishchenko *et al.* 2018).

Assuming that the principal-value problem has a simple real-frequency solution, $\mathcal{L}_R(\omega_r)\varphi = 0$, and that the resonant part is a small correction, we write $\omega = \omega_r - i\gamma$, with $\gamma > 0$ for damping of the amplitude, and project. Since the kinetic operator is local in ψ (there is no radial drift) and $\mathcal{N}_a$ is real at real frequency, $\mathcal{L}_R$ is symmetric under $\int d^3\mathbf{r}$, and its left null vector is φ itself. Hence

$$\gamma = \frac{\langle \varphi, \mathcal{L}_I(\omega_r)\varphi \rangle}{\langle \varphi, \partial_\omega \mathcal{L}_R(\omega_r)\varphi \rangle} = \frac{\pi T}{2n_0 e^2} \frac{\sum_a e_a \int d^3\mathbf{r} \int d^3\mathbf{v} \mathcal{N}_a(\omega_r) |\bar{\varphi}|^2 \delta(\omega_r - \Omega_a)}{\langle \varphi, \partial_\omega \mathcal{L}_R(\omega_r)\varphi \rangle}, \quad (4.8)$$

where $\langle f, g \rangle \equiv \int d^3\mathbf{r} f^* g$, and all explicit frequency dependence, including that in $\mathcal{N}_a$, is differentiated in the denominator. For a scalar dispersion relation $\mathcal{D}(\omega) = 0$, (4.8) reduces to $\gamma = \mathcal{D}_I/\mathcal{D}'_R$. The sign of the numerator is not fixed: the resonance can damp or drive, and negligible net injection is not a proof that $\gamma > 0$. Recall from §2.1 that $\Gamma_E = 2\gamma$ for a structure of fixed shape.

Equation (4.8) requires three caveats, each of which turns out to matter. Firstly, in a pair plasma whose energy-dependent drifts have opposite signs for the two species, the resonant sets $kU + k\bar{v}_{dp}[0, \infty)$ and $kU + k\bar{v}_{de}[0, \infty)$ together cover the whole real frequency axis. There is therefore never a discrete real spectrum: the ‘mode’ of (4.8) is always a quasi-mode on the second Riemann sheet, and the expansion is valid only when $|\gamma| \ll |\omega_r - kU|$, equivalently $|\mathcal{D}_I| \ll |\tilde{\omega} \mathcal{D}'_R|$, and only if the expansion connects to an actual second-sheet root. A real root of $\mathcal{D}_R$ is not sufficient (§4.3). Secondly, freezing the zonal state requires the quasi-mode frequency $|\tilde{\omega}|$ to greatly exceed both the rate of zonal evolution and the breathing frequency ω_b ; otherwise the relevant object is a Floquet mode of a time-periodic background, not (4.8). Thirdly, (4.8) describes neither a strongly damped transient nor a finite-amplitude structure; the latter is the subject of §5.

The resonance condition $\omega_r = kU(\psi) + k\bar{v}_{da}$ can be satisfied in two quite different ways, and the rest of this section treats them in turn. If the zonal flow is locally uniform across the vortex and the vortex moves relative to it, the resonance selects an energy, and the damping is a genuine Landau damping on the magnetic drift (§4.3). If, instead, the drift is negligible compared with the variation of kU across the vortex, the resonance selects a radius, viz., a critical layer where $U = \omega_r/k$ (§4.4); a weak drift then acts as a perturbation localised at that layer (§4.5).

4.3. Limit (a): uniform flow and drift resonance

Assuming that (i) the vortex is flute-like, $\bar{\varphi} = \varphi$; (ii) U is locally uniform, so the zonal flow enters only as the Doppler shift of (4.4); (iii) the drift is isotropic in pitch angle, $k\bar{v}_{dp} = \hat{\omega}_d u$ and $k\bar{v}_{de} = -\hat{\omega}_d u$, with $u \equiv \varepsilon/T$ and $\hat{\omega}_d$ the signed drift frequency of a positron at $u = 1$; and (iv) the background, including the zonal flattening, is a local Maxwellian with species-symmetric effective gradients, $k\tilde{v}_{*p}(u) = -k\tilde{v}_{*e}(u) = -\hat{\omega}_d \Omega_* [1 + \eta(u - 3/2)]$, the dispersion relation can be written in closed form. The energy

measure at fixed pitch angle is

$$p(u) = \frac{2}{\sqrt{\pi}} u^{1/2} e^{-u}, \quad \int_0^\infty p(u) du = 1, \quad (4.9)$$

and the sign of Ω_* is chosen so that $\Omega_* > 0$ is the destabilising side.[†] With $z \equiv \tilde{\omega}/\hat{\omega}_d$ and $\kappa \equiv k_\perp \lambda_D$, (4.5) reduces to

$$\mathcal{D}(z) = 1 + \kappa^2 - \left\langle \frac{z^2 - u \Omega_* [1 + \eta(u - 3/2)]}{z^2 - u^2} \right\rangle_u = 0, \quad (4.10)$$

where $\langle \cdot \rangle_u \equiv \int_0^\infty p(u) (\cdot) du$ with the causal continuation. The function $\mathcal{D}$ is even in $\tilde{\omega}$, so the quasi-modes of a single uniform flow come in pairs $\pm \tilde{\omega}$ that damp at the same rate. This degeneracy does not by itself explain the equal decay of the two simulated vortices, which ride on different jets (§6.1); that follows from the species-exchange symmetry of §2.2. The exact evaluation in terms of the plasma dispersion function (Fried & Conte 1961) is given in Appendix A. The resonant part, for $\hat{\omega}_d > 0$ and $z > 0$, is

$$\mathcal{D}_I = \sqrt{\pi} z^{1/2} e^{-z} \left[z - \Omega_* \left(1 + \eta \left(z - \frac{3}{2} \right) \right) \right]. \quad (4.11)$$

For $z \gg 1$, the principal-value part has the asymptotic expansion

$$\mathcal{D}_R = \kappa^2 - \sum_{n \geq 0} \frac{m_n}{z^{2n+2}}, \quad m_n = \langle u^{2n+2} \rangle - \Omega_* \left[\left(1 - \frac{3}{2} \eta \right) \langle u^{2n+1} \rangle + \eta \langle u^{2n+2} \rangle \right], \quad (4.12)$$

where $\langle u^m \rangle = \Gamma(m + 3/2)/\Gamma(3/2)$, so that $m_0 = 15/4 - (3/2)\Omega_*(1 + \eta)$. There is thus a weakly damped branch with

$$z_r^2 \approx \frac{m_0}{\kappa^2} + \frac{m_1}{m_0}, \quad \gamma \approx |\hat{\omega}_d| \frac{\sqrt{\pi}}{2} \frac{z_r^{3/2} e^{-z_r}}{\kappa^2} \left[z_r - \Omega_* \left(1 + \eta \left(z_r - \frac{3}{2} \right) \right) \right], \quad (4.13)$$

where $\mathcal{D}'_R \approx 2\kappa^2/z$ has been used. Equation (4.13) implies that the resonance damps if and only if $z_r > \Omega_*[1 + \eta(z_r - 3/2)]$, which, at large z_r , reduces to $\Omega_*\eta < 1$ for the effective (flattened) gradients. For a pure density gradient, or for profiles flattened by the condensate, the resonance always damps; for $\Omega_*\eta > 1$, the high-energy resonant particles drive a growing kinetic mode, with growth rate of order $\hat{\omega}_d$.[‡]

The damping (4.13) is linear in the drift frequency $\hat{\omega}_d$, and its dependence on wavenumber,

$$\gamma \propto k_y k_\perp^{-9/2} \exp\left(-\frac{\sqrt{m_0}}{k_\perp \lambda_D}\right), \quad (4.14)$$

has an essential singularity at $k_\perp \rightarrow 0$: long-wavelength vortices are exponentially protected in this limit, and at $k_y = 0$ the channel vanishes altogether. A subthreshold increase of Ω_* lowers m_0 , hence z_r , and so increases the damping; conversely, flattening of the profiles by the condensate reduces it. If the drift depends on pitch angle, the class with the largest $|\bar{v}_d|$ has the smallest z_r and dominates the exponential.

Table 2 lists exact second-sheet roots of (4.10). The projection (4.8), i.e., $\mathcal{D}_I/\mathcal{D}'_R$

[†] The sign convention is fixed by the Z-pinch check. With the anisotropic drift $\propto v_\parallel^2 + v_\perp^2/2$ in place of assumption (iii), the fluid limit of the model reproduces the interchange threshold $\Omega_*(1 + \eta) = 7/4$ of Mishchenko *et al.* (2018) and Kennedy & Plunk (2026b) only if $\Omega_* > 0$ on the destabilising side. In the isotropic model, the corresponding fluid threshold is $\Omega_*(1 + \eta) = 5/2$.

[‡] Exact second-sheet roots confirm this: $\Omega_* = 1.2$, $\eta = 1$ gives $\gamma/\hat{\omega}_d \approx -1.3$ at $\kappa = 0.1$, and $\Omega_* = 0.8$, $\eta = 2$ gives $\gamma/\hat{\omega}_d \approx -2.7$. Note that this flute-like isotropic model has no unstable root at $\Omega_* = 1$, $\eta = 0$, and so does not contain the entropy mode of Kennedy & Plunk (2026b); this is a limitation of the model, not a statement about the real system.

$k_{\perp}\lambda_D$	$\Omega_* = 0, \eta = 0$	$\Omega_* = 0.5, \eta = 0$	$\Omega_* = 1, \eta = 0$	$\Omega_* = 1, \eta = 1$
0.05	4.1×10^{-11}	1.7×10^{-9}	1.1×10^{-7}	2.9×10^{-4}
0.1	3.7×10^{-4}	1.9×10^{-3}	1.1×10^{-2}	0.10
0.15	2.9×10^{-2}	6.9×10^{-2}	0.16	—
0.2	0.16	0.27	0.45	—

TABLE 2. Damping rate $\gamma/\hat{\omega}_d$ of the weakly damped quasi-mode of the uniform-flow model (4.10), from exact second-sheet roots. For $\Omega_* = 0.5, \eta = 1$, the corresponding values are 6.5×10^{-8} at $k_{\perp}\lambda_D = 0.05$ and 7.2×10^{-3} at 0.1. The closed-form estimate (4.13) tracks z_r to about 0.2% for $\Omega_* \leq 0.5$ and γ to about 10% at $k_{\perp}\lambda_D = 0.1$.

evaluated at the exact real root of $\mathcal{D}_R$, agrees with the exact second-sheet damping rate of table 2 to better than 1% for $\kappa \leq 0.1$, to 2–10% at 0.15–0.2, and to about 25% at 0.3, beyond which the quasi-mode is strongly damped and (4.8) is not usable. The closed-form asymptotic estimate (4.13) is cruder, agreeing to about 10% at $\kappa = 0.1$; e.g., for $\Omega_* = 0$ it gives 4.0×10^{-4} against the exact 3.7×10^{-4} . Real roots of $\mathcal{D}_R$ also exist elsewhere — at $z \approx 2.4$, with $\mathcal{D}'_R < 0$, and, for $\Omega_* = 1, \eta = 0$, at $z \approx 0.25$, with $\mathcal{D}'_R > 0$ and $\mathcal{D}_I < 0$, suggesting a spurious ‘growth’ of $0.26\hat{\omega}_d$ — but neither has a complex root near it, which illustrates the first caveat of §4.2.

Weak radial variation of U can be included at leading order, provided that there is no critical layer, i.e., $|\omega - kU(\psi)| \gg |\hat{\omega}_d|$ everywhere across the vortex. Expanding about the vortex centre, we find that the resonant energy becomes $z_r(\psi) \approx z_{r0} - (U'/\bar{v}_{d1})\delta\psi - (U''/2\bar{v}_{d1})\delta\psi^2$, where $\bar{v}_{d1}$ is the drift at $u = 1$ and the wavenumber cancels. For a Gaussian $|\varphi|^2$ of radial variance σ^2 , linear shear then multiplies the damping by $\exp[(U'\sigma/\bar{v}_{d1})^2/2]$, always increasing it; the control parameter is the velocity jump across the vortex compared with the thermal drift speed. At a jet maximum, $U' = 0$, the factor becomes $(1 - sU''\sigma^2/\bar{v}_{d1})^{-1/2}$, with $s = \text{sgn } \tilde{\omega}$ for $\hat{\omega}_d > 0$, so the curvature of the jet breaks the degeneracy of the $\pm\tilde{\omega}$ partners. We have not tested these corrections numerically.

Finally, let us note what this limit requires. The weakly damped branch has $z_r \approx \sqrt{m_0}/k_{\perp}\lambda_D \gg 1$, viz., the vortex must move through the local zonal flow much faster than the thermal drift speed, and the frozen-background caveat of §4.2 requires $|\hat{\omega}_d|z_r \gg \omega_b$. A vortex that co-moves with the flow somewhere inside itself, $\tilde{\omega}(\psi) = 0$, is not this quasi-mode at all. We now consider that case.

4.4. Limit (b): no drift, the Euler equation and the critical layer

Suppose instead that the vortex is flute-like and that the magnetic drift is negligible compared with the variation of the Doppler-shifted frequency across it, $|k\bar{v}_{da}| \ll |k\delta U|$. Then the kinetic problem collapses to a fluid one, as the following argument shows.[†]

Let us take the charge moment $\sum_a e_a \int d^3\mathbf{v}$ of (2.2) with $\bar{\phi} = \phi$. The bracket commutes with the velocity integral, the gradient drive cancels between the species because $e_a(e_a f_0/T)v_{*a}^T = -e_a \partial_{\psi} f_0$ sums to zero, and the drift term survives only through the even part $h_p + h_e$, since $e_a \bar{v}_{da}$ is the same for both species. Using (2.3)

[†] This reduction does not appear in the earlier note.

to write the charge moment as $(2n_0e^2/T)(1 - \lambda_D^2 \nabla_\perp^2)\phi$, and $\{\phi, \phi\} = 0$, we find[‡]

$$\partial_t \nabla_\perp^2 \phi + \{\phi, \nabla_\perp^2 \phi\} = \frac{T}{2n_0e\lambda_D^2} \partial_\alpha \int d^3\mathbf{v} \bar{v}_{dp} (h_p + h_e), \quad (4.15)$$

where $\bar{v}_{dp}$ is the positron drift. Without the drift, (4.15) is the two-dimensional Euler equation for the streamfunction ϕ and the vorticity $\nabla_\perp^2 \phi$: the Debye length drops out, the gradients drop out, as they do for a symmetric pair plasma in a straight field (Helander 2014), and the vortex-jet system is, at this order, an incompressible inviscid flow. Linearising about $\Phi(\psi)$ with $\phi = \bar{\phi} + \varphi e^{ik\alpha - i\omega t}$, and writing $c \equiv \omega/k$, gives Rayleigh's equation (Lord Rayleigh 1880; Drazin & Reid 2004),

$$(U - c) (\partial_\psi^2 - k^2) \varphi - U'' \varphi = 0, \quad (4.16)$$

in which the zonal vorticity gradient U'' plays the role of the velocity gradient of the distribution function in Landau damping. The same equation follows from the kinetic response: setting $\bar{v}_{da} = 0$ in (4.2) and (4.4) and using the zonal part of (2.3), $\sum_a e_a \int d^3\mathbf{v} \delta F_a = -(2n_0e^2\lambda_D^2/T) U'$, the operator (4.5) becomes λ_D^2 times the left-hand side of (4.16) divided by $c - U$, which is a useful check on both.

The resonance of (4.7) is now the critical layer $U(\psi_c) = c$, and the projection (4.8) becomes the classical critical-layer (Landau-type) damping of a Rayleigh quasi-mode (Briggs *et al.* 1970),

$$\gamma = \pi \frac{\sum_c U_c'' |\varphi_c|^2 / |U_c'|}{-\partial_{\omega_r} \text{PV} \int d\psi k U'' |\varphi|^2 / (\omega_r - kU)}, \quad (4.17)$$

where $k > 0$ (otherwise a factor $\text{sgn } k$ appears), the sum runs over the critical layers, the subscript c denotes evaluation there, and the integral is over ψ . The derivative must be taken outside the principal value: written as $\int k U'' |\varphi|^2 / (\omega_r - kU)^2 d\psi$, the denominator has a double pole at each critical layer, which a principal value does not regularise, and must be read as a Hadamard finite part. The same caveats as for (4.8) apply, and so does the remark about its sign: where the zonal vorticity gradient U_c'' at the critical layers has the other sign, (4.17) is negative and the mode grows, which is Rayleigh's shear-flow instability.[†] In this limit, the magnetic drift and the kinetic structure of h_a are corrections, of relative size $|\bar{v}_{da}|/|U - c|$, that shift the critical layer by an energy-dependent distance $\delta\psi \sim \bar{v}_{da}/|U'|$. Crucially, the critical layer is a property of the jet profile, not of the plasma's velocity distribution; its nonlinear theory is that of the Kelvin cat's eye (Stewartson 1978; Warn & Warn 1978), with viscosity (Haberman 1972) in place of collisions.

Which of the two limits applies is a matter of measurement: the vortex speed relative to the local zonal flow must be compared with the thermal drift speed. We shall see in §6.1 that the simulated vortices are deep in limit (b): the variation of the zonal flow across each vortex exceeds the drift speed by orders of magnitude, and $U = c$ occurs inside each vortex.

[‡] Strictly speaking, (4.15) holds as written in a flux tube whose metric is constant along the field line; in the dipole, $\nabla_\perp^2$ depends on l , and a flute-like ϕ satisfies it only on average along the line. This is consistent with the approximate flute-like character of the long-wavelength states (Kennedy & Plunk 2026a). In the flux tube of the simulations, the average is weighted by the Jacobian, and its effect on (4.16) is to replace k^2 in the Laplacian by $k^2 \langle g^{yy} \rangle / \langle g^{xx} \rangle$, where g^{xx} and g^{yy} are the metric coefficients; this correction is of order unity, and the stability of the jets depends on it (§6.2).

[†] Divided by $U - c$, (4.16) reads $(\partial_\psi^2 - k^2)\varphi + kU''\varphi/(\omega - kU) = 0$. Equation (4.17) follows from regularising $1/(\omega - kU + i0)$ by (4.6), multiplying by φ^* and integrating, which gives $\mathcal{D}_I = -\pi \sum_c U_c'' |\varphi_c|^2 / |U_c'|$. It is evaluated for the jets of the simulation in §6.2.

4.5. Weak drift on a sheared flow: resonance at the critical layer

Between the two limits lies the case that will concern us in §6: a flute-like vortex with critical layers, as in limit (b), and a drift that is small but not zero. The right-hand side of (4.15) is then the only term that can change the electrostatic energy of the vortex other than its exchange with the zonal flow. Multiplying (4.15) by $-\phi$ and integrating gives

$$\frac{d}{dt} \int \frac{|\nabla_{\perp} \phi|^2}{2} = - \int \phi \{ \phi, \nabla_{\perp}^2 \phi \} - \frac{T}{2n_0 e \lambda_D^2} \int \phi \partial_{\alpha} \int d^3 \mathbf{v} \bar{v}_{dp} (h_p + h_e), \quad (4.18)$$

where the integrals are over ψ and α ; the first term on the right-hand side is the exchange between the vortex and the zonal flow, which cancels in the total, and the second is the work done on the particles through their drift. Let us evaluate the latter with the linear response (4.2) and the prescription (4.6). The positrons resonate where $\omega - kU(\psi) = k\bar{v}_{dp}$ and the electrons where $\omega - kU(\psi) = -k\bar{v}_{dp}$, i.e., on opposite sides of each critical layer and at a distance $\bar{v}_{dp}/|U'_c|$ from it that depends on energy. Integrating across the layer, over which φ hardly varies, we find that the drift converts the electrostatic energy $E_v = \int d\psi (|\partial_{\psi} \varphi|^2 + k^2 |\varphi|^2)/4$ of the vortex at the rate

$$\Gamma_d \equiv - \frac{1}{E_v} \left(\frac{dE_v}{dt} \right)_{\text{drift}} = \frac{2\pi}{k\lambda_D^2} \langle \omega_d (\omega_d - \omega_*) \rangle \frac{\sum_c |\varphi_c|^2 / |U'_c|}{\int d\psi (|\partial_{\psi} \varphi|^2 + k^2 |\varphi|^2)}, \quad (4.19)$$

where $\omega_d \equiv k\bar{v}_{dp}$, $\omega_* \equiv -k\bar{v}_{*p}$ is the diamagnetic frequency built from the gradients at the layer, the angle brackets denote the average over the Maxwellian, and the sum is over the critical layers. In the model of §4.3, $\langle \omega_d (\omega_d - \omega_*) \rangle = \hat{\omega}_d^2 m_0$, with the same m_0 as in (4.12).

Three properties of (4.19) are worth noting. Firstly, it is independent of the amplitude of the vortex, as any linear rate must be. Secondly, it is of second order in the drift, but it contains no exponentially small factor: the shear allows particles of every energy to find their resonance, each at its own radius, so the protection of long wavelengths found in §4.3 is lost. Thirdly, its two terms have different meanings: $\langle \omega_d^2 \rangle$ is a conversion of electrostatic energy into entropy, which always damps, while $\langle \omega_d \omega_* \rangle$ is the injection of free energy by the gradients, which drives. Below the fluid interchange threshold, the sum damps. Note that (4.19) assumes unperturbed orbits in a layer whose width, $\bar{v}_{dp}/|U'_c|$ per unit of thermal energy, is a small fraction of the width of the cat's eye of any vortex of interest (§5.2); whether it survives trapping is, therefore, precisely the question, to which we return in §6.8.

4.6. Passive phase mixing

For comparison, let us consider a passive perturbation obeying $\partial_t g + i(kU + \hat{\omega}_d u)g = 0$, initially proportional to the Maxwellian. Its normalised moment is

$$C(t) = e^{-ikUt} \int_0^{\infty} p(u) e^{-i\hat{\omega}_d u t} du = e^{-ikUt} (1 + i\hat{\omega}_d t)^{-3/2}, \quad (4.20)$$

$$|C(t)|^2 = (1 + \hat{\omega}_d^2 t^2)^{-3/2},$$

whose logarithmic decay rate, $3\hat{\omega}_d^2 t / (1 + \hat{\omega}_d^2 t^2)$ for $|C|^2$ (half of it for $|C|$), first rises and then falls as $3/t$. Phase mixing alone therefore produces an algebraic tail that decelerates, not a decay that accelerates (see also Schekochihin *et al.* 2016 on the competition between linear phase mixing and nonlinear advection). The same is true of the self-consistent response: $\mathcal{K}_0$ of Appendix A has a branch point at $\tilde{\omega} = 0$, the zero-energy particles, so even the full response has an algebraic tail alongside the Landau pole. Note, however,

that (4.20) is the law of limit (a), i.e., phase mixing by the energy-dependent drift. In limit (b), passive mixing is instead by the shear of the jet across the critical layer (Orr-type mixing), whose algebraic laws differ and which we have not worked out for the simulated profile. Equation (4.20) is the basis for the test of the late remnant in §6.9, which therefore probes drift phase mixing only.

5. Nonlinear theory: trapping and sinks

The linear response of §4 ignores the modification of trajectories by the vortex itself. In this section, we estimate that modification, first for the geometry in which the critical layer lies on the flank of a jet, then for the jet maximum assumed in the earlier note, and finally ask how trapping interacts with the hyperdiffusive sinks.

5.1. The moving-frame Hamiltonian

We use local coordinates (x, y) , radial and binormal, and the $\mathbf{E} \times \mathbf{B}$ streamfunction Ψ .[†] In the frame $\xi = y - ct$ moving with a nearly steady vortex, and approximating $\bar{v}_{da}$ as radially constant, the trajectories of an orbit class satisfy

$$\dot{x} = -\partial_\xi \mathcal{H}, \quad \dot{\xi} = \partial_x \mathcal{H}, \quad \mathcal{H} = \Psi_{\text{zon}}(x) + \Psi_v(x, \xi) + (\bar{v}_{da} - c)x, \quad (5.1)$$

so closed orbits must be sought in the moving, orbit-dependent Hamiltonian, not in the instantaneous laboratory-frame streamlines. When $|\bar{v}_{da}| \ll |U - c|$ across the vortex, as in limit (b) of §4.4, the drift term is a small correction and the trapped region is the fluid cat's eye at $U = c$, shared by all energy classes.

5.2. Shear at the critical layer: the pendulum

Let x_r satisfy $U(x_r) + \bar{v}_{da} = c$, with $U = \Psi'_{\text{zon}}$ and $S \equiv U'(x_r) \neq 0$. If the radial variation of the vortex is negligible over the trapped region, expansion of (5.1) gives the pendulum Hamiltonian

$$\mathcal{H} = \text{const} + \frac{S}{2}X^2 - A \cos(k\xi) + \dots, \quad X = x - x_r, \quad (5.2)$$

where A is the vortex amplitude in streamfunction units. The small-orbit frequency and the separatrix half-width are

$$\omega_{\text{tr}} = |k| \sqrt{|S|A}, \quad w = 2\sqrt{\frac{A}{|S|}}, \quad (5.3)$$

the latter from equating $SX^2/2$ to the barrier $2A$; both scale as $A^{1/2}$. For $S > 0$, the centre is at $X = \xi = 0$; for $S < 0$, it is shifted by half a wavelength. More generally, at an elliptic fixed point of the full $\mathcal{H}$,

$$\omega_{\text{tr}}^2 = \det \nabla \nabla \mathcal{H} = \mathcal{H}_{xx} \mathcal{H}_{\xi\xi} - \mathcal{H}_{x\xi}^2 > 0, \quad (5.4)$$

which includes the radial curvature of the vortex.

[†] In GENE units, $\Psi = \phi/C_{xy}$ with $C_{xy} = 0.9264$ the metric factor of the flux tube, x is in ρ_{ref} and t in $L_{\text{ref}}/c_{\text{ref}}$. GENE writes the real field as $\phi = \phi_0 + 2\text{Re}(\phi_{k_y} e^{ik_y y})$, so the cosine amplitude of a single k_y is $A = 2|\phi_{k_y}|/C_{xy}$; using $|\phi_{k_y}|$ instead would underestimate the trapping frequency and width below by $\sqrt{2}$ each.

5.3. The jet maximum

The earlier note located the vortices at the maxima of the zonal flow, where $U' = 0$ and (5.2) is singular. For completeness, let us work this case out, although we shall find in §6.1 that it is not the geometry of the simulation. Expanding about the maximum x_m , with mismatch $\delta \equiv U(x_m) + \bar{v}_{da} - c$ and vortex $A g(X) \cos k\xi$, where $g(0) = 1$, $g'(0) = 0$ and $\kappa_\phi^2 \equiv -g''(0)$, we obtain

$$\mathcal{H} = \delta X + \frac{U''}{6} X^3 + A g(X) \cos k\xi. \quad (5.5)$$

At exact resonance, $\delta = 0$, there are two cases. If the vortex is radially curved enough, $\kappa_\phi^3 A \gg |U''|$, the vortex makes its own cell, with $\omega_{tr} = |k| \kappa_\phi A$, linear in A , closing for $|X| < 2A\kappa_\phi^2/|U''|$. If the vortex is radially flat, $\kappa_\phi^3 A \ll |U''|$, there are no closed orbits at all, since $\dot{\xi} = U''X^2/2$ never changes sign, and the nonlinear scales are those of a non-twist map, viz., a width $(A/|U''|)^{1/3}$ and a frequency $\omega_{NL} = |k| |U''|^{1/3} A^{2/3}$. For $\delta \neq 0$, twin island chains appear at $X_* = \pm(2|\delta|/|U''|)^{1/2}$, with local shear $|U''|X_*$, and reconnect when their widths become comparable with X_* . In other words, at a jet maximum, $\omega_{tr} \propto A$ or $A^{2/3}$, not $A^{1/2}$.

5.4. Trapping, the O'Neil threshold and the sinks

Following O'Neil (1965), trapping modifies a resonant response before linear damping removes the perturbation if $\omega_{tr} \gtrsim \gamma$. On the flank, (5.3) turns this into a critical amplitude

$$A_{crit} \sim \frac{\gamma^2}{k^2 |S|}, \quad (5.6)$$

where γ is the amplitude damping rate; at a jet maximum, the corresponding estimates are $A_{crit} \sim \gamma/|k| \kappa_\phi$ and $A_{crit} \sim \gamma^{3/2}/(|k|^{3/2} |U''|^{1/2})$. In O'Neil's picture, $A \ll A_{crit}$ gives linear damping and $A \gg A_{crit}$ gives damping that is suppressed after a few trapping periods. This picture predicts that the decay should slow down as the amplitude grows, and speed up as it falls — qualitatively what is observed — but it says nothing about how fast the suppressed state decays.

That rate is set by the sinks. In the trapped regime, the comparison that matters is not ω_{tr} against γ but ω_{tr} against the effective rate at which the sinks erode the structure inside the cat's eye. For an order- n hyperdiffusion of coefficient ν_n acting across a trapped region of half-width w , the relevant parameter, analogous to that of viscous critical layers (Haberman 1972), is

$$\Lambda \equiv \frac{\nu_{eff}}{\omega_{tr}} = \frac{2\nu_n}{k|S|w^{n+1}} \propto A^{-(n+1)/2}. \quad (5.7)$$

For $\Lambda \gg 1$, the sinks destroy the cat's eye faster than it forms, and the damping is linear, $\gamma = \gamma_L$; for $\Lambda \ll 1$, trapping suppresses it, and $\gamma \approx C\gamma_L \Lambda^q$ with some C and q that depend on the sink. A Krook-type sink gives $q = 1$, whence $\gamma \propto A^{-5/2}$ for $n = 4$. A separatrix boundary layer of thickness $\delta_b = (\nu_n/\omega_{tr})^{1/n}$, across which the trapped and passing populations are reconnected, suggests instead $\gamma/\gamma_L \sim \delta_b/w \propto \Lambda^{1/n}$, i.e., $\gamma \propto A^{-5/8}$ for $n = 4$; this is a heuristic, not a derivation. In either case, the decay accelerates as A falls, without any crossover at $\omega_{tr} \sim \gamma$.

5.5. Phase mixing feeding a sink

A second elementary calculation separates the transfer of free energy to fine scales from its removal. Near a resonance, let the orbit frequency vary linearly with a local velocity

offset v , $\Omega(v) \approx \Omega_r + \beta v$, and, removing the constant phase, consider

$$\partial_t f + i\beta v f = -\nu_4 \partial_v^4 f, \quad \nu_4 > 0. \quad (5.8)$$

With $\hat{f}(\kappa_v, t) \equiv \int e^{-i\kappa_v v} f dv$, this becomes $\partial_t \hat{f} - \beta \partial_{\kappa_v} \hat{f} = -\nu_4 \kappa_v^4 \hat{f}$, whose characteristics are $\kappa_v(t) = \kappa_{v0} - \beta t$. Along the one launched at $\kappa_{v0} = 0$,

$$\hat{f}(-\beta t, t) = \hat{f}(0, 0) \exp\left(-\frac{\nu_4 \beta^4 t^5}{5}\right), \quad t_{\text{sink}} \sim (\nu_4 \beta^4)^{-1/5}, \quad (5.9)$$

since $\int_0^t (\kappa_{v0} - \beta s)^4 ds = [\kappa_{v0}^5 - (\kappa_{v0} - \beta t)^5]/5\beta$. Mixing makes the velocity-space wavenumber grow as $|\beta|t$, and the fourth-order sink then acts as κ_v^4 . Note that (5.9) describes a moving Fourier component of the distribution, not $E_{\text{nz}}(t)$, and that it shows that accelerating removal of fine structure can occur without any trapping transition: acceleration alone does not identify the mechanism.

The same law, with $\beta \rightarrow k_y S$ and $v \rightarrow x$, describes the shearing of open-streamline structure in physical space by a perpendicular hyperdiffusion. Interpreted as a radial width, t_{sink}^{-1} then corresponds to the hyperviscous critical-layer width $\delta_\nu = (\nu_4/k|S|)^{1/5}$. Whether either version applies in the simulation depends on whether the relevant wavenumber is resolved before t_{sink} , to which we return in §6.8.

6. Confrontation with the simulations

We now test the calculations of §4 and §5 against the reduced-box simulation and its restarts (§2.3). All measurements are made at the outboard mid-plane, and, as in §2.2, $k_y = 1$ denotes the vortex and $k_y = 0$ the jet. The two counter-propagating vortices are separated by a two-frequency least-squares fit, $a_+ e^{-i\omega_+ t} + a_- e^{-i\omega_- t}$, in windows of about sixteen breathing periods, whose residual is small until shortly before the collapse.

6.1. Where the vortices sit and which limit applies

The geometry is not the one assumed in the earlier note [see figure 4(a)]. Each vortex propagates in the direction of its own jet, at a fixed fraction of the peak speed of the jet, which changes only during the relaxation that follows the burst [see figure 5(c) for the frequency]. Each vortex is radially broad — a plateau over its own jet with maxima on both flanks — and its maxima lie just inside the critical layers $U = c$, where the shear is near its largest. In the frame moving with a vortex, the flow at the jet maximum is therefore far from stagnant, so the jet maximum is not an $\mathbf{E} \times \mathbf{B}$ fixed point and (5.5) does not describe the simulation: the pendulum case of §5.2 applies. The streamlines of the moving-frame Hamiltonian (5.1) show the cat's eyes directly, one on each flank of the jet, staggered by half a wavelength because the shear has opposite signs on the two flanks [see figure 4(b)].

The comparison that decides between the limits of §4 is as follows. At the wavenumber of the vortex, the drift frequency of a thermal particle and the diamagnetic frequency of the background are nearly equal,[†] and both are smaller than the Doppler frequency of the vortex relative to its jet, $k_y(U_{\text{jet}} - c)$, by orders of magnitude. The energy-dependent shift of the resonant layer, $\delta x \sim \bar{v}_{da}/|S|$, is therefore a small fraction of a grid cell, whereas the trapped half-width is several grid cells in the slow phase of the decay and still about one at the collapse. The vortices are thus in limit (b), the drift resonance is in effect

[†] Here the magnetic drift is the one that the code applies, viz., the curvature and ∇B drift of its geometry, averaged along the field line.

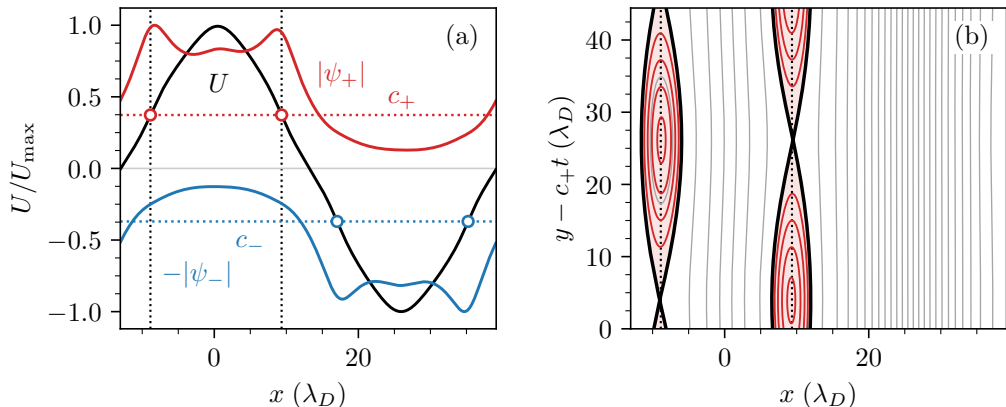


FIGURE 4. Geometry of the vortices on the zonal jets, at one time in the slow phase of the decay. (a) The zonal flow U (black), normalised to its maximum, and the radial profiles of the two vortices, $|\psi_+|$ (red) and $-|\psi_-|$ (blue), each normalised to its own maximum; the horizontal dotted lines are the phase speeds $c_{\pm}$ of the two vortices in the same normalisation, the open circles mark the critical layers $U = c_{\pm}$, and the vertical dotted lines mark those of the $+$ vortex. (b) Streamlines of the moving-frame Hamiltonian (5.1) of the $+$ vortex, with $\bar{v}_{da} = 0$ and the $-$ vortex left out (grey); the separatrices are drawn in heavy black, and the trapped orbits inside them in red. The radial coordinate is shifted by a quarter of the box so that neither jet is cut by the periodic boundary. Each vortex sits on its own jet, its critical layers lie on the flanks of that jet, not at its maximum, and a cat's eye several Debye lengths wide forms at each of them.

an $\mathbf{E} \times \mathbf{B}$ critical layer, and the uniform-flow results of §4.3, including the exponential protection (4.14), do not apply to them.

Placing the vortices in limit (b) does not, however, remove the frozen-background caveat of §4.2. The condition $|\hat{\omega}_d|z_r \gg \omega_b$ of limit (a) would fail here, since ω_b exceeds every drift frequency in the problem; but in limit (b), too, the Doppler frequency of the vortex relative to its jet, $k_y(U_{\text{jet}} - c)$, is somewhat below ω_b , and it vanishes at the critical layer. The quasi-mode is therefore not fast compared with the breathing in either limit, and freezing the zonal state, on which (4.16) and (4.17) rest, is justified only if the breathing modulates U by a small relative amount. It does: the modulation of U at the breathing frequency is of the order of $E_{\text{nz}}/E_{\text{zon}}$, i.e., small from the slow phase of the decay onwards, and it displaces the critical layers by a small fraction of a grid cell. The frozen zonal state is thus a fair premise for the linear problem.

The two vortices decay at nearly the same rate in the original run and in both amplitude tests [the band of figure 7(a) spans the two]. This is expected in either geometry and does not discriminate between them. The vortices ride on different jets and are related by the species-exchange symmetry of §2.2; even at a jet maximum, the curvature factor $(1 - sU''\sigma^2/\bar{v}_{d1})^{-1/2}$ of §4.3 would be the same for both, since U'' and the sign s of $\tilde{\omega}$ both reverse between the jets. The asymmetry of §4.3 is between the $\pm\tilde{\omega}$ partners on the same jet.

6.2. The vortices as the shear instability of their jets

Let us now solve the problem of §4.4 for the jets of the simulation, i.e., Rayleigh's equation (4.16), with the field-line-averaged metric, on the measured zonal profile. It has no weakly damped quasi-mode. Instead, it has one unstable mode in each direction, whose frequency follows that of the vortices throughout the run, while both change substantially, and whose radial structure is that of the vortices [see figure 5(b,c)]. The

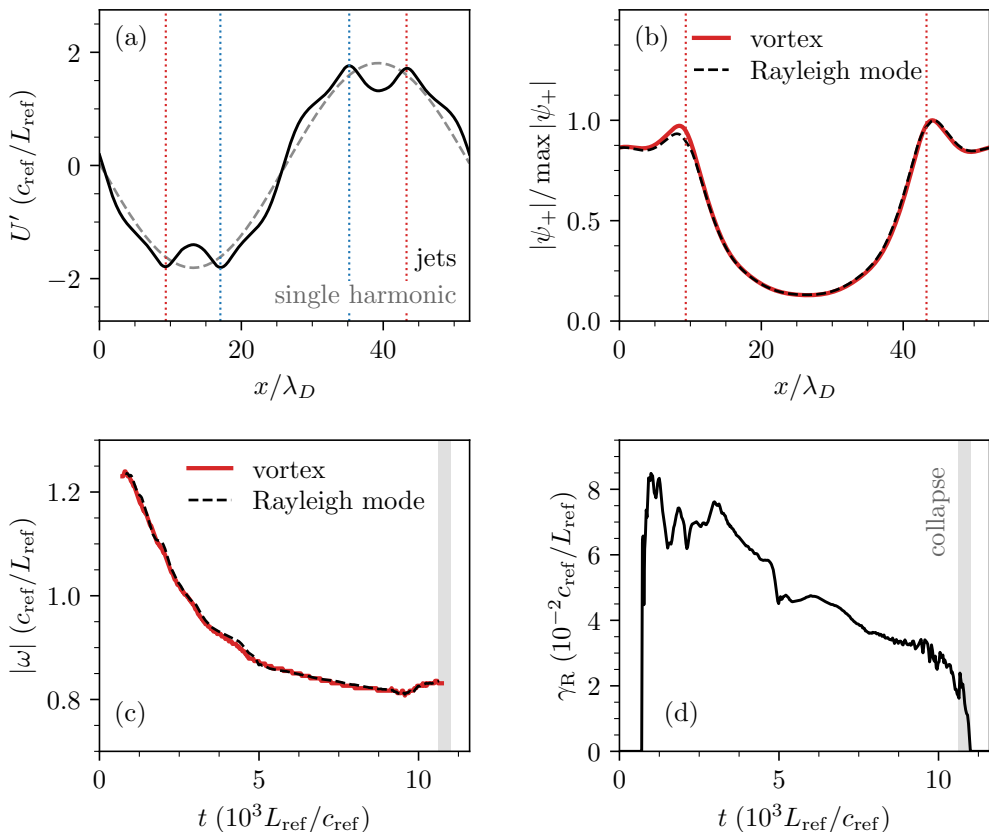


FIGURE 5. The vortices as the shear-flow instability of the jets. (a) The zonal vorticity U' of the measured jets (black) and of a single harmonic of the same amplitude (grey dashed), at one time in the slow phase of the decay; the vertical dotted lines mark the critical layers $U = \pm c$ of the vortex + (red) and - (blue). (b) The radial structure of the vortex + (red) and of the unstable mode of Rayleigh's equation (4.16) on the same jets (black dashed). (c) The frequency of the vortices (red) and of the unstable Rayleigh mode on the measured jets (black dashed) through the run. (d) The growth rate γ_R of that mode; the grey band marks the collapse. The vortices are the unstable shear mode of jets whose shear layers have split into vorticity strips, and they disappear when that instability does.

reason is visible in the zonal vorticity: each shear layer between the jets is split into two strips, so that U'' changes sign close to each critical layer, and the critical layers sit on the strips [see figure 5(a)]. A single harmonic of the same amplitude is stable; it is the corrugation of the jets at the Debye scale that makes them unstable. The formula (4.17) gives the sign and, closely, the size of the growth rate γ_R .

In other words, the vortices are not damped waves that happen to survive: they are the saturated shear-flow instability of their own jets. This explains at once why the jets supply them with energy (§6.8), and why a vortex that is scaled down on an unchanged zonal profile regrows (§6.4): it is a small perturbation of an unstable jet.

It also makes a prediction for their amplitude. An instability of a shear flow that grows at the rate γ_R saturates when the fluid elements at its critical layers are trapped, i.e., when ω_{tr} is a multiple of order unity of γ_R (O'Neil 1965), and it stays at that level if γ_R

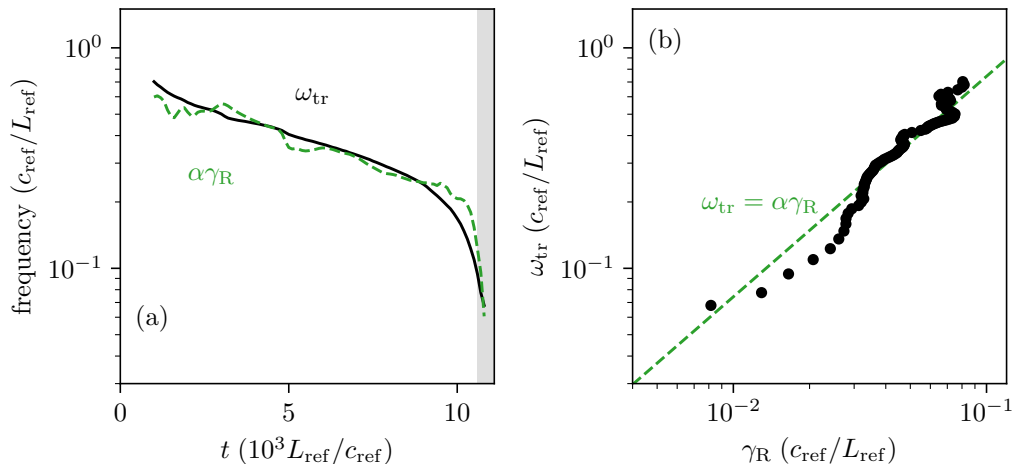


FIGURE 6. The saturation law (6.1). (a) The trapping frequency ω_{tr} of the vortices, from (5.3) at their critical layers (black), and the growth rate γ_{R} of the unstable Rayleigh mode of the measured jets, multiplied by the constant α (green dashed), both as means over about eighty breathing periods; α is the geometric mean of their ratio, and the grey band marks the collapse. (b) The same data, ω_{tr} against γ_{R} , with the line $\omega_{\text{tr}} = \alpha\gamma_{\text{R}}$. The amplitude of the vortices follows the growth rate of the instability of their jets: the vortices decay because the jets become stable.

changes slowly. Hence, by (5.3),

$$\omega_{\text{tr}} = \alpha \gamma_{\text{R}}, \quad \text{i.e.,} \quad A = \frac{\alpha^2 \gamma_{\text{R}}^2}{k^2 |S|}, \quad E_{\text{nz}} \propto \gamma_{\text{R}}^4, \quad (6.1)$$

with α a constant. Figure 6 tests this, with ω_{tr} measured from the vortices and γ_{R} computed from the jets, two independent inputs. The law holds through the whole decay, including the collapse, with a single value of α , while the energy of the vortices falls by orders of magnitude. The growth rate vanishes at the collapse [see figure 5(d)].

The decay of the vortices is thus slaved to the slow loss of the instability of the jets, and the question of why the vortices decay becomes the question of what erodes the vorticity strips. Two processes do: the perpendicular hyperdiffusion, which acts on the fine zonal structure directly, and the vortices themselves, which draw their energy from the strips at the rate at which they are damped (§6.8).

6.3. Trapping against damping

At the critical layer, the pendulum estimate (5.3) gives a trapping frequency that falls steadily, as $A^{1/2}$, over the decay [see figure 7(a)]; the Hessian (5.4) at the numerically located O-point agrees with it closely. The ratio $\omega_{\text{tr}}/\gamma_{\text{amp}}$ is very large in the slow phase of the decay, falls as the decay accelerates, and is still large at the collapse; the amplitude tests lie in the same range.

The O’Neil criterion is therefore satisfied by a wide margin throughout, and the crossover $\omega_{\text{tr}} \sim \gamma$ is never reached: the critical amplitude (5.6), evaluated with the measured damping rate, is smaller than the amplitude at the critical layer by orders of magnitude even at the collapse. Therefore, the acceleration of the decay is not the transition from trapping-suppressed to linear damping that motivated the earlier note. The trapped regime does hold throughout, which makes the sink-limited regime of (5.7) the natural replacement hypothesis; we shall see in §6.8 that it does not survive either, because the damping turns out not to depend on amplitude at all.

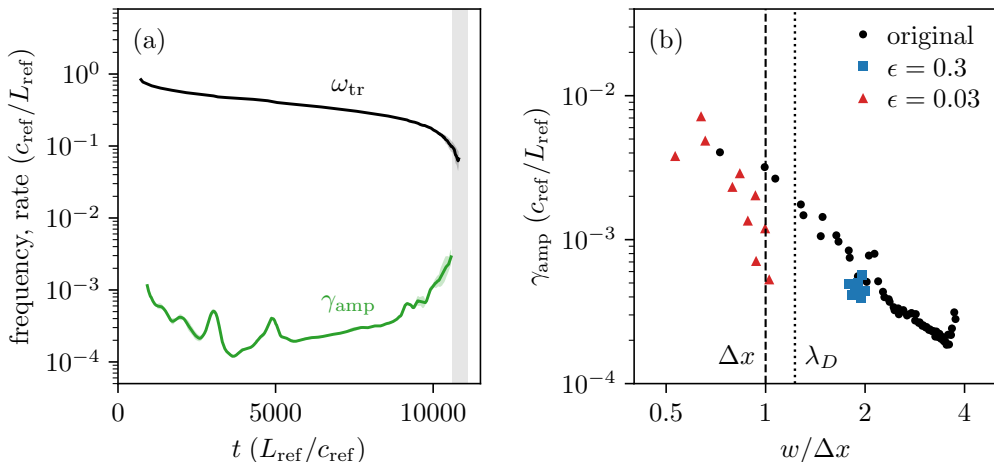


FIGURE 7. Trapping against damping. (a) The pendulum trapping frequency ω_{tr} of (5.3) at the critical layer (black) and the amplitude damping rate γ_{amp} (green), each as the geometric mean of the energy of each vortex in windows of about a hundred breathing periods, and the grey band marks the collapse. (b) The damping rate of each vortex, from fits in windows half as long, against the half-width w of its cat’s eye in units of the grid spacing, for the original run in the slow phase of the decay and the collapse (black dots) and for the $\epsilon = 0.3$ (blue squares) and $\epsilon = 0.03$ (red triangles) restarts after their initial transient; the dashed and dotted vertical lines mark $w = \Delta x$ and $w = \lambda_D$. The trapping frequency exceeds the damping rate by orders of magnitude at all times, and the damping rate rises steeply as the cat’s eye narrows towards one grid cell, which is also about one Debye length.

Two further facts complicate the picture. Firstly, the breathing frequency exceeds ω_{tr} even in the slow phase, so the cat’s eye is periodically modulated at a frequency faster than its own orbits, and the trapped orbits near the separatrix may well be chaotic; we have not analysed this. Secondly, both (5.6) and (5.7) presuppose a positive linear damping rate γ_L , whereas the linear mode on these jets grows (§6.2); the trapped state is that of a saturated instability, not of a damped wave.

6.4. Amplitude dependence

The amplitude tests exclude the linear null hypothesis decisively [see figure 3(a)]. When the breathing is removed, either by a least-squares fit of $\ln E_{\text{nz}}$ to a line plus two harmonics of ω_b or by a linear fit to a running mean over ten breathing periods, the decay rates of the original, the $\epsilon = 0.3$ and the $\epsilon = 0.03$ runs over their common window increase in that order, and the two methods agree; smaller amplitude means faster decay. The restarts also violate linear scaling more directly: immediately after the restart, the rescaled energy E_{nz}/ϵ^2 grows, the more so the smaller ϵ . This transient is the instability of §6.2: the rescaling leaves the zonal profile untouched, so the smaller vortex finds itself on an unstable jet and grows, at a rate comparable with the Rayleigh rate, until it is trapped again [see figure 13(b)].

At matched amplitude, the decay rate is approximately, but not exactly, a function of the amplitude alone [see figure 3(b)]. As the original run decays through the amplitudes at which the restarts were started, its decay rate rises to values close to theirs, and all three runs follow roughly one curve. History matters, however: at matched amplitude, the restarts carry appreciably less entropy per unit electrostatic energy than the original

[see figure 9(b)]. A fit to the natural decay of the original run gives $\Gamma_E \propto A_c^{-p}$, where A_c is the amplitude at the critical layer, with p of order unity, somewhat smaller early in the decay and somewhat larger late.

Figure 3(b) also shows the predictions of §5.4. The linear prediction, viz., a rate independent of amplitude, and the Krook-type law, $\Gamma_E \propto A^{-5/2}$, are excluded, and the data lies between the boundary-layer heuristic, $p = 5/8$, and $p = 1$. Taken by itself, this would suggest a sink-limited trapped regime. It should not be read in that way: the budget of §6.8 shows that the net decay rate is the difference of two rates, of which only one depends on amplitude, and that one is not a dissipation rate.

6.5. The collapse and the grid

The one measured quantity that tracks the decay rate across all three runs is the separatrix half-width w of (5.3). It is several grid cells in the slow phase of the decay and falls through one grid cell as the vortices collapse; the decay rate rises steeply as w approaches Δx and keeps rising through the collapse, with no plateau [see figure 7(b)]. The $\epsilon = 0.3$ restart lies on the rate-against-width curve of the original. The $\epsilon = 0.03$ restart starts nominally below one grid cell, regrows until $w \approx \Delta x$, and then decays at a rate that starts below that of the original at the same width and rises more steeply. The continuum hyperviscous layer width of §5.5, $\delta_\nu = (\nu_4/k|S|)^{1/5}$ with $\nu_4 = \nu_x(\Delta x/2)^4$, is a fraction of one grid cell, so the effective dissipative width at the critical layer is the grid itself. This suggests that the collapse occurs when the cat's eye can no longer be resolved.

There is, however, a confounding coincidence: $\lambda_D \approx 1.2 \Delta x$, so under-resolution of the cat's eye cannot be separated from physics at the Debye scale in this simulation.† A restart from the accelerating phase of the decay with twice the radial resolution decides between the two: if the collapse is set by the grid, it should move to an amplitude about four times lower, i.e., an E_{nz} about sixteen times lower, while a Debye-scale effect would not move. This restart, together with four others in which one of the sinks is changed by a factor of four, is running at the time of writing (§2.3). Finally, let us note that the collapse rate is not the linear rate of the original vortex: after the collapse, the state is different (§6.9), so the collapse rate cannot be read as γ_L .

6.6. Flute character and the Debye scale

The reduction of §4.4 assumed a flute-like vortex. This holds for the vortex as a whole, but not scale by scale (see figure 8). The part of the $k_y = 1$ potential that varies along the field line is a negligible fraction of its power at the box scale, grows steeply with the radial wavenumber, and saturates at a value of order unity for $k_x \lambda_D \gtrsim 1$, in the slow phase of the decay and before the collapse alike; and the share of the vortex power that resides at these scales increases as the collapse approaches. This is what (2.3) requires: since $k_\perp$ varies along the field line, a flute-like potential cannot be sustained once $k_\perp \lambda_D \sim 1$ (Kennedy & Plunk 2026b). The minimal estimate, in which the right-hand side of (2.3) is taken to be constant along the field line, so that $\varphi \propto (1 + k_\perp^2 \lambda_D^2)^{-1}$ with the $k_\perp$ of the flux tube, reproduces the dependence on wavenumber, including the saturation at the Debye scale, but underestimates the non-flute power considerably [see figure 8(b)]: the charge response of the plasma must itself vary along the field line, as it does when the orbit average (2.1) depends on pitch angle.

This has three consequences for the present problem. Firstly, the critical layer, whose

† Recall from §4.4 that λ_D drops out of the drift-free vortex dynamics, which favours the former; but the flute-like approximation behind (4.15) itself degrades at $k_\perp \lambda_D \sim 1$.

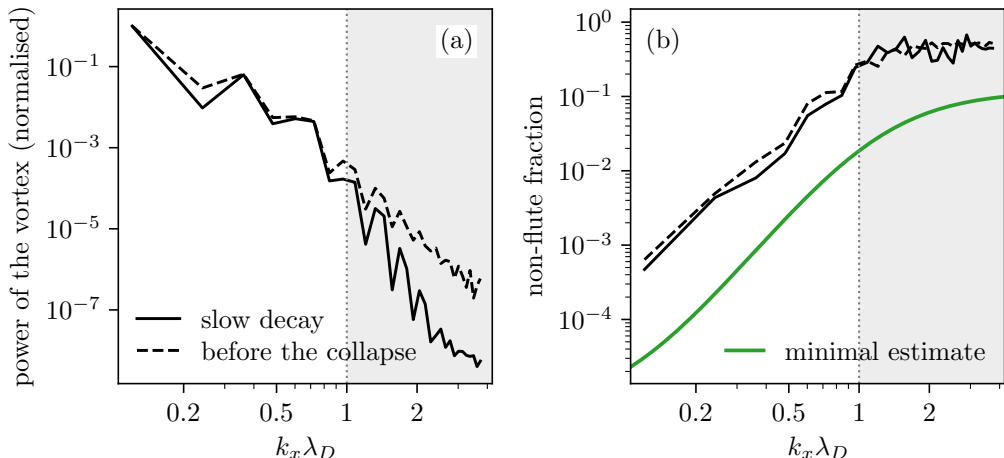


FIGURE 8. Flute character of the vortex, scale by scale. (a) The power of the $k_y = 1$ potential, summed along the field line and averaged over all output times in the slow phase of the decay (solid line) and in the approach to the collapse (dashed line), against the radial wavenumber $k_x \lambda_D$, each normalised to its maximum. (b) The fraction of that power that is not flute-like, i.e., the power of the departure of the potential from its mean along the field line divided by the total, for the same two intervals, and the minimal estimate $\varphi \propto (1 + k_{\perp}^2 \lambda_D^2)^{-1}$ (green). The vertical dotted line marks $k_x \lambda_D = 1$, and scales smaller than the Debye length are shaded. The vortex is flute-like at the box scale and loses its flute character at the Debye scale.

width is of the order of w , is governed by (4.15) only while $w \gg \lambda_D$, whereas, in the simulation, w falls from a few Debye lengths to less than one. Secondly, a potential that is not flute-like has an orbit average (2.1) that depends on pitch angle, so particles of different pitch angle are advected through the cat's eye differently. This is the phase mixing in pitch angle that carries entropy to small scales of velocity space in the turbulent state (Kennedy & Plunk 2026b). Thirdly, a potential that varies along the field line does work on particles streaming along it, which is a route from electrostatic energy to entropy that a flute-like vortex does not have; we shall see in §6.8 that this is how the vortices are damped. The coincidence of §6.5 thus has a physical candidate on the Debye side: the collapse may mark the amplitude at which the cat's eye has shrunk to the scale where the vortex ceases to be flute-like.

6.7. Entropy and free energy of the vortices

If the decay were passive phase mixing, the electrostatic energy lost by the vortices would reappear, at least for a while, as entropy, and W_{nz} would be approximately conserved while E_{nz} fell. The ratio Z_{nz}/E_{nz} does rise, slowly during the decay and steeply through the collapse [see figure 9]. However, Z_{nz} decays almost as fast as E_{nz} through most of the decay, so W_{nz} falls at nearly the same rate as either; only in the final collapse does Z_{nz} clearly lag behind. The amplitude tests behave alike.

In other words, the W_{nz} -conserving passive picture does not describe the decay: the free energy leaves the non-zonal part as a whole. Where it goes, and where it comes from, is the subject of the next subsection.

6.8. The energy budget of the vortices

The totals E_{nz} and W_{nz} say how fast the vortices decay, but not why. The free-energy diagnostics of GENE (Bañón Navarro *et al.* 2011) resolve every term of the energy balance

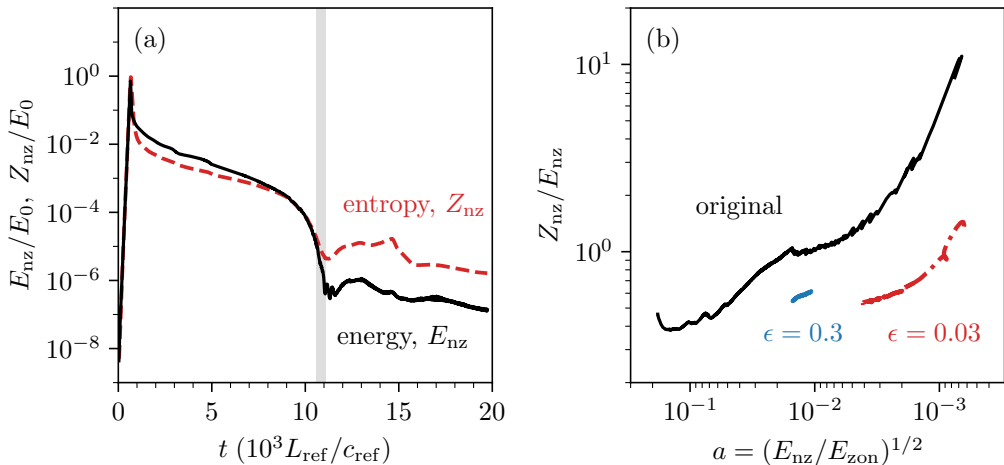


FIGURE 9. Entropy of the vortices. (a) The non-zonal electrostatic energy E_{nz} (black solid) and entropy Z_{nz} (red dashed) of the original run, as running means over about ten breathing periods, in units of E_0 ; their sum is the non-zonal free energy W_{nz} . (b) The ratio Z_{nz}/E_{nz} against the amplitude measure a , decreasing to the right, for the original run up to the end of the collapse (black solid) and for the two restarts after their initial transient (blue dashed, red dash-dotted). The entropy of the vortices falls together with their energy until the collapse, so their free energy is not conserved, and at the same amplitude the restarts carry less entropy than the original: the state of the vortices is not a function of their amplitude alone.

in (k_x, k_y) and separate the electrostatic energy from the entropy, so the budget of the electrostatic energy of the non-zonal part can be written by itself,

$$\frac{dE_{nz}}{dt} = T - L, \quad (6.2)$$

where T is the nonlinear transfer of electrostatic energy from the zonal flow to the vortices, i.e., the average over the breathing of the exchange (2.5), L is the rate at which the linear terms of the kinetic equation, viz., parallel streaming and the magnetic drift, convert the electrostatic energy of the vortices into entropy, and the direct action of the perpendicular sink on E_{nz} , which is negligible until the collapse, has been dropped.[†] Dividing by E_{nz} turns each term into a rate. Three facts emerge (see figure 10).

Firstly, the vortices are not merely damped. Throughout the slow phase of the decay, the jets supply them with electrostatic energy at a rate larger than the net decay rate, so the slow decay of figure 2 is the small difference between two larger rates: the breathing exchange has a mean as well as an oscillating part, and the mean is directed from the jets to the vortices, as it must be if the vortices are an instability of the jets (§6.2). This is also why the zonal flow decays faster while the vortices are large than after they have gone [see figure 2(b)].

Secondly, the loss rate L/E_{nz} is constant while the energy of the vortices falls by orders of magnitude, and it is the same in the amplitude tests as in the original run. In other words, the loss is linear in the energy of the vortices. The entropy so produced is removed by the velocity-space hyperdiffusion, which accounts for most of the free energy removed

[†] All rates in this subsection are averages over the breathing, which the output resolves with only about three samples per period; the mean is fitted together with two harmonics of ω_b , and the scatter between consecutive windows gives the errors shown in figure 10. The budget of the free energy W_{nz} , in which the loss term is the removal by the sinks and the transfer is obtained as a residual, leads to the same three conclusions.

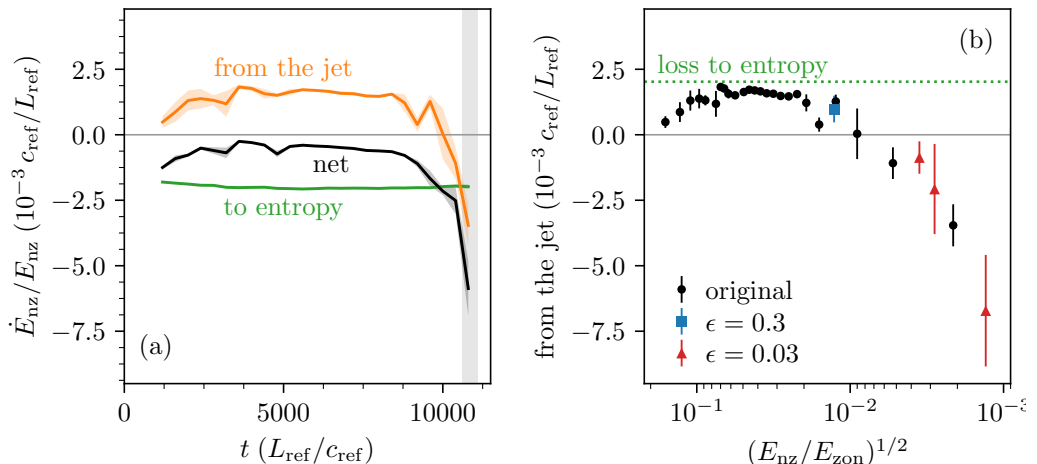


FIGURE 10. The budget (6.2) of the electrostatic energy of the vortices, averaged over the breathing. (a) The rates of change of E_{nz} , per unit E_{nz} , due to the nonlinear exchange with the zonal flow (orange), due to the linear terms, which convert electrostatic energy into entropy (green), and their sum (black); the bands are standard errors over consecutive windows, and the grey band marks the collapse. (b) The nonlinear rate against the amplitude measure $a = (E_{nz}/E_{zon})^{1/2}$, which decreases to the right, for the original run (black dots) and for the $\epsilon = 0.3$ (blue square) and $\epsilon = 0.03$ (red triangles) restarts; the dotted line is the magnitude of the linear loss. The vortices lose electrostatic energy to entropy at a steady rate that does not depend on their amplitude; the jets replace most of it while the vortices are large, and the decay accelerates because that supply fades and finally reverses.

from the non-zonal part [see figure 11(b)]; the perpendicular sink, by contrast, acts almost entirely on the zonal flow, at a rate that the vortices do not affect [see figure 11(a)].

Thirdly, the amplitude dependence of the decay resides entirely in the exchange with the jets. As the vortices weaken, the supply rate T/E_{nz} stays nearly constant for a long time, then falls, passes through zero, and becomes a drain: small vortices give energy back to the zonal flow [see figure 10(b)]. The restarts follow the same curve at a fixed time: the $\epsilon = 0.3$ restart has the loss rate of the original and a smaller supply, and the $\epsilon = 0.03$ restart has the loss rate of the original and, once its initial regrowth is over, a drain. The acceleration of the decay is therefore the fading, and finally the reversal, of the transfer from the jets, not an increase of the damping. Only in the last stage before the collapse does the removal of free energy itself speed up, as the perpendicular sink begins to act on the vortices directly.

This changes the question. What damps the vortices is linear; what is nonlinear is their exchange with the jets. Let us confront each with the calculations of §4 and §5.

The linear loss is not the drift resonance (see figure 12). With the measured profiles of the vortices, the measured shear at their critical layers and the drift of the code, (4.19) predicts that the drift should convert the electrostatic energy of the vortices into entropy faster than the whole of the measured loss, and that the gradients should inject energy at a comparable rate. Neither happens: the conversion through the drift is far smaller than this prediction, and the injection is negligible. This is as it should be. The resonant layers of (4.19) lie deep inside cat's eyes that are far wider than they are, where the particles are trapped and circulate at a frequency that far exceeds the drift frequency, so the unperturbed-orbit resonance does not exist. In other words, the drift resonance is

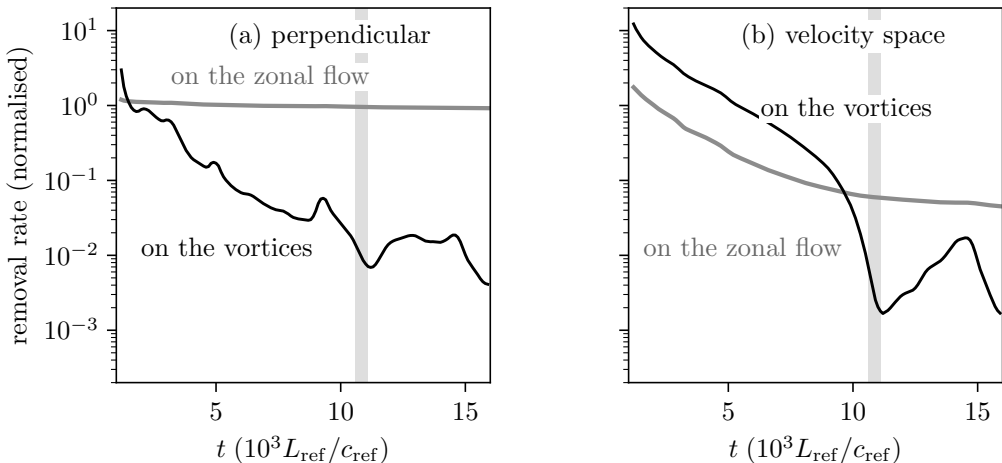


FIGURE 11. Where the free energy is removed. The rate of removal by (a) the perpendicular and (b) the velocity-space hyperdiffusion, each split into the part that acts on the zonal ($k_y = 0$; grey) and on the non-zonal ($k_y \geq 1$; black) part of the distribution, in sliding windows of about a hundred breathing periods, in units of the perpendicular removal from the zonal part in the slow phase of the decay; the parallel sink is negligible and is not shown. The grey band marks the collapse. The perpendicular sink belongs to the zonal flow and the velocity-space sink to the vortices.

suppressed by trapping, as O’Neil (1965) would have it, and it stays suppressed down to the collapse.

Instead, nearly all of the linear loss is in the parallel channel: it is work done on particles streaming along the field line by the part of the vortex potential that varies along it, i.e., Landau damping of the non-flute part of the vortices, whose fine structure in $v_{\parallel}$ is what the velocity-space sink then removes. A loss of this kind is linear, since the field equation (2.3) makes the non-flute part proportional to the flute-like one (§6.6), and it is not affected by trapping in the cat’s eyes, which concerns the motion across the field. The transit-averaged equations of §2.1 do not contain it: they assume that particles sample the whole field line many times in one period of the perturbation, whereas the Doppler-shifted frequency of the vortex over most of its width is not small compared with the transit frequency. An estimate of its rate is $L/E_{\text{nz}} \sim 2\gamma_{\parallel}f$, where f is the fraction of the power of the vortices that resides in the non-flute part and $\gamma_{\parallel}$, of the order of the transit frequency, is the rate at which that part is Landau damped; with the measured f (figure 8), this is of the right order of magnitude. Since f increases steeply with $k_{\perp}\lambda_D$, the loss should be weaker in a larger box, and it is [see figure 12(b)]. A calculation of f and $\gamma_{\parallel}$ from first principles is the obvious next step; we have not done it.

As for the exchange, it is the instability of §6.2 seen in the energy budget. While the jets are unstable, the vortices draw on them, at the rate at which they are damped and no faster, since their amplitude is held at the saturated level (6.1); as the instability fades, so does the supply; and when the jets have become stable, the sign of the exchange is that of critical-layer absorption (§4.4), viz., the vortices give up their energy to the mean flow and collapse. O’Neil’s picture — absorption at small amplitude, suppressed by trapping at large amplitude — thus does describe the exchange between the vortices and the jets, although not with the threshold of (5.6). The supply fades and reverses as the trapped half-width approaches one grid cell, i.e., about one Debye length (§6.5 and §6.6).

Finally, let us ask how much of this the fluid limit contains by itself. We integrate

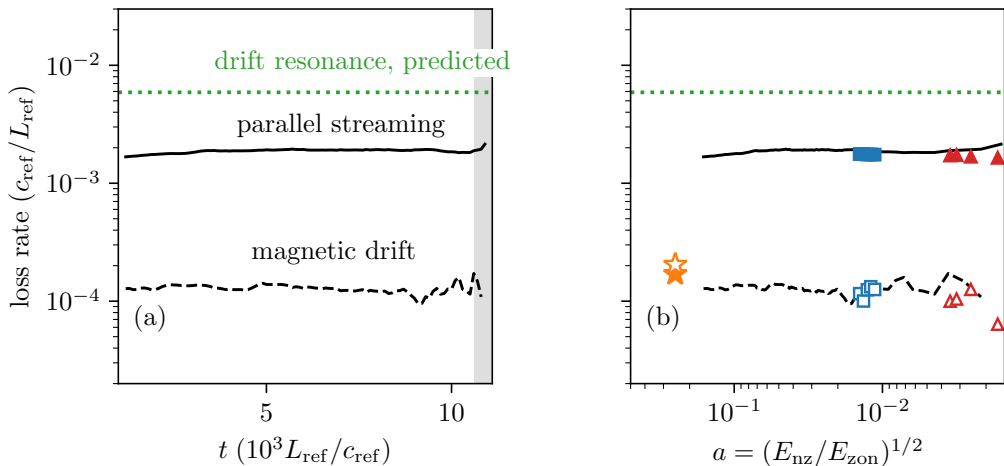


FIGURE 12. The channels through which the vortices lose electrostatic energy to entropy. (a) The rate of loss of E_{nz} , per unit E_{nz} , through parallel streaming (black solid) and through the magnetic drift (black dashed), in sliding windows of about a hundred breathing periods of the original run, and the prediction (4.19) for the drift channel, evaluated for the vortices of figure 4 with the gradients at the critical layers taken to be flattened (green dotted); the grey band marks the collapse. (b) The same two rates against the amplitude measure $a = (E_{nz}/E_{zon})^{1/2}$, which decreases to the right, for the original run (lines) and the $\epsilon = 0.3$ (squares) and $\epsilon = 0.03$ (triangles) restarts, and for the production run of Kennedy & Plunk (2026b), whose box is four times larger in each perpendicular direction (stars). The loss is carried by parallel streaming, at a rate that does not depend on amplitude; the drift resonance, which the linear calculation puts above it, is suppressed.

the two-dimensional Euler equation in the box of the simulation, with the same Fourier modes and with its perpendicular hyperdiffusion,[†] starting from the potential of the simulation in the slow phase of the decay (see figure 13). The fluid model reproduces the two counter-propagating vortices, their phase speed, the breathing, and the regrowth of the amplitude tests. Without the sink, its vortices do not decay; with it, they decay at a rate roughly proportional to the sink, faster at smaller amplitude, and they collapse once small enough. However, the fluid model decays too slowly, and the reduced restart regrows in it at the full Rayleigh rate, faster than in the simulation. The damping that the fluid limit lacks is the kinetic loss L .

The restarts of §2.3 bear on these conclusions: if the kinetic loss is Landau damping, it should not depend on the coefficient of the velocity-space hyperdiffusion; if the erosion of the vorticity strips is due to the perpendicular hyperdiffusion, the decay should speed up and slow down with it; and if the loss of the instability is set by the grid, the collapse should move to smaller amplitudes when the radial resolution is doubled.

6.9. The late remnant

The passive drift-mixing law (4.20) of limit (a) predicts an algebraic tail, $|C|^2 \propto t^{-3}$, once phase mixing has run its course. After the collapse, E_{nz} does not decay cleanly (see

[†] The perpendicular hyperdiffusion of the simulation acts on h_a , whose charge moment is proportional to $(1 + k_{\perp}^2 \lambda_D^2) \phi$, so in the fluid limit it damps the vorticity at its nominal rate multiplied by $1 + 1/k_{\perp}^2 \lambda_D^2$. With this sink and no other, the fluid model reproduces the measured decay rate of the zonal flow, which is therefore the perpendicular hyperdiffusion acting on the Boltzmann part of the distribution.

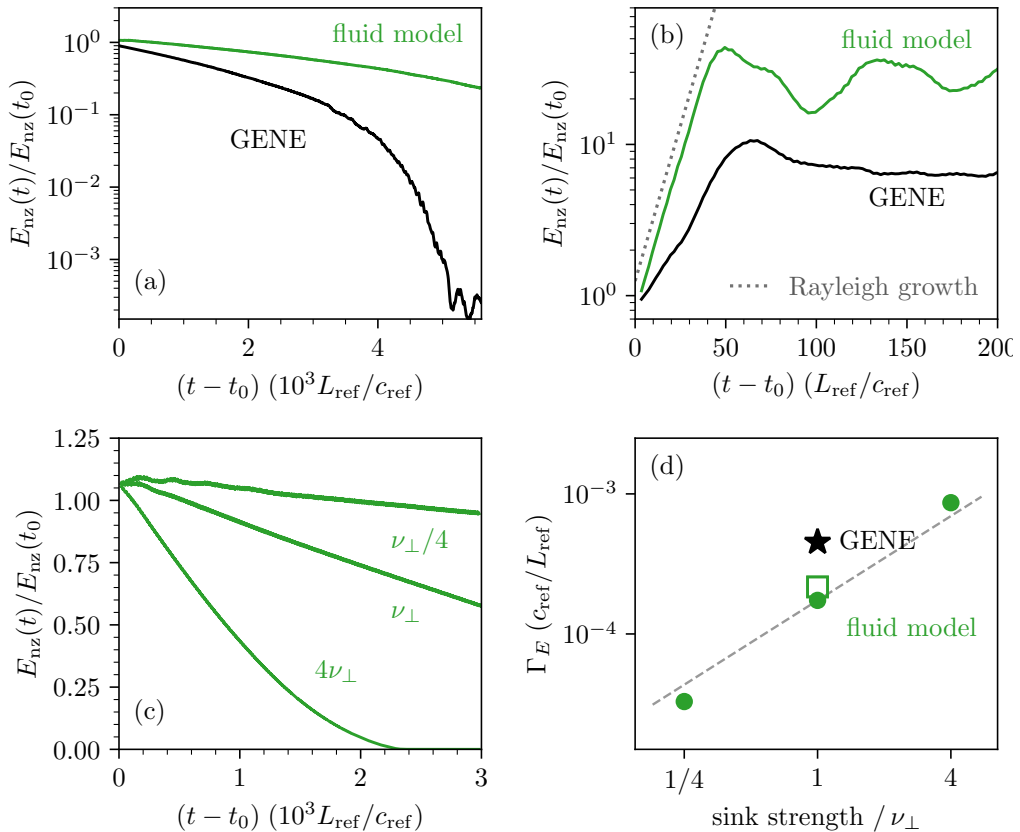


FIGURE 13. The two-dimensional fluid model against the kinetic simulation. (a) The non-zonal energy of the simulation (black) and of the fluid model started from its potential at the time t_0 of the amplitude tests (green). (b) The same for the restart with the vortices reduced by $\epsilon = 0.03$, with growth at the Rayleigh rate (dotted). (c) The fluid model with its perpendicular sink multiplied by four, unchanged, and divided by four. (d) The decay rate of the fluid model against the strength of its sink (circles; the square is at twice and four times the radial resolution), a linear dependence (dashed) and the simulation (star). All curves are running means over ten breathing periods [two in (b)]. A fluid with the perpendicular sink of the simulation reproduces the vortices, their regrowth and a part of their decay; the rest of the damping is kinetic.

figure 14): it regrows once and then falls slowly and irregularly. A law $(t - t_c)^{-3}$ measured from the time t_c of the collapse would require a far steeper fall and is excluded. Power laws measured from the start of the run or from the burst fit as well as an exponential and, over this window, cannot be distinguished from one.

The remnant is almost entirely $k_y = 1$, and its entropy greatly exceeds its electrostatic energy, but the coherent two-vortex fit explains only part of it, so it is partly incoherent, and the passive law may not apply to it in any case. Its phase speed is also lower than that of the original vortices and keeps falling, and its trapped half-width is well below one grid cell. Its budget is also different in kind: after the collapse, the rates at which free energy is supplied to the non-zonal part by the zonal flow and removed from it by the sinks are both several times larger than before it and nearly equal, with most of the removal by the perpendicular sink, so the remnant is not a freely decaying vortex but a forced response to the zonal state. We therefore draw no conclusion from it beyond

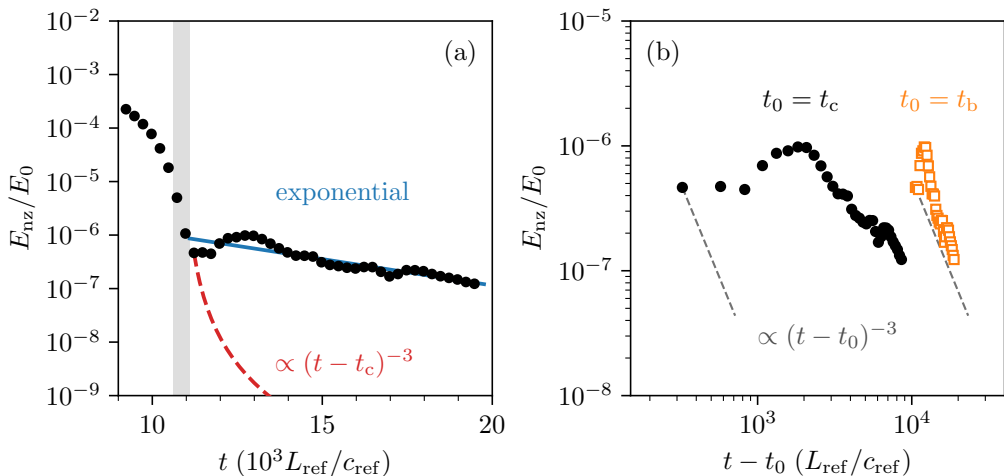


FIGURE 14. The remnant after the collapse. (a) The non-zonal electrostatic energy E_{nz}/E_0 as geometric means over blocks of about seventy breathing periods (dots), with an exponential fitted to the remnant (blue solid) and the passive drift-mixing law $(t - t_c)^{-3}$ of (4.20), measured from the collapse time t_c and anchored at the first block of the remnant (red dashed). (b) The same block means against $t - t_0$ on logarithmic axes, with $t_0 = t_c$ (black dots) and $t_0 = t_b$, the time of the burst (orange squares); the dashed lines have slope -3 . What is left after the collapse decays slowly and irregularly, far more slowly than a t^{-3} tail started at the collapse.

excluding a t^{-3} tail from the collapse. Since the vortices are in limit (b), that exclusion tests only drift phase mixing, not the shear (Orr-type) phase mixing at the critical layer that would be the relevant passive mechanism (§4.6).

7. Summary and discussion

Starting from the drift-resonance picture of the earlier note, we have derived the linear response of a vortex on a frozen zonal condensate, corrected it, completed it in two limits, and confronted it, together with simple trapping and sink models, with a reduced-box simulation and restarts of it. Let us state what we consider established.

Firstly, the corrected linear response (4.4) is Doppler invariant only once the Boltzmann part of the zonal distribution is retained, and its drive is the gradient of the distribution as modified by the condensate. In a pair plasma the drift resonances of the two species cover the whole real frequency axis, so the weak-damping formula (4.8) always describes a second-sheet quasi-mode, valid only for $|\gamma| \ll |\omega_r - kU|$. In the uniform-flow limit, the resonance damps if and only if $z_r > \Omega_*[1 + \eta(z_r - 3/2)]$, the rate is linear in the drift frequency, and long-wavelength vortices are exponentially protected, (4.14). Secondly, in the drift-free limit, the charge moment of the transit-averaged equations reduces, in a field-line-averaged sense, to the two-dimensional Euler equation (4.15), the vortex obeys Rayleigh's equation (4.16), and the resonance is a hydrodynamic critical layer.

Thirdly, a weak drift on a sheared flow resonates at the critical layers, where it converts the electrostatic energy of a vortex into entropy at the rate (4.19), which is of second order in the drift, independent of amplitude and free of the exponential protection of the uniform-flow limit.

Fourthly, the simulated vortices are in the drift-free limit as far as their motion is concerned: they propagate at a fixed fraction of the jet speed, their critical layers lie on the flanks of the jets at near-maximal shear, with resolved cat's eyes, and the magnetic

drift shifts the resonance by a small fraction of the width of the trapped region. They are, moreover, the unstable mode of Rayleigh’s equation on the measured jets, which predicts their frequency throughout the run and their radial structure, and their amplitude obeys the saturation law (6.1) while their energy falls by orders of magnitude. The vortices are the saturated shear-flow instability of jets whose shear layers are corrugated at the Debye scale.

Fifthly, the damping of the vortices is linear and kinetic. Their electrostatic energy is converted into entropy at a rate that does not depend on their amplitude, and almost all of this conversion is Landau damping along the field line of the part of the vortices that is not flute-like; the entropy is then removed by the velocity-space sink. The drift resonance is not the mechanism: it is far smaller than (4.19) predicts, because its resonant layers lie inside the cat’s eyes, where the particles are trapped. The jets, meanwhile, feed the vortices at nearly the rate at which they are damped, as the instability requires.

Finally, the vortices decay, with an amplitude dependence that excludes any linear explanation, because the jets slowly lose their instability; they collapse when the jets become stable and the exchange with them reverses. What is left afterwards is a forced response to the zonal state, not a decaying vortex, which is why it does not follow the t^{-3} drift-mixing tail of (4.20). A two-dimensional fluid with the perpendicular sink of the simulation reproduces the vortices, their regrowth after a reduction of their amplitude, and a part of their decay, but not the kinetic loss.

The following remain open. We have no first-principles calculation of the kinetic loss: the minimal estimate of the non-flute part of the vortices from (2.3) has the right dependence on wavenumber but is too small, and the Landau damping of that part is outside the transit-averaged equations. Nor do we have a theory of the constant α of (6.1), or of the rate at which the vorticity strips of the jets are eroded, which, with (6.1), would give the lifetime of the vortices. It remains an open question what sets the amplitude at which the jets become stable: it coincides with the trapped half-width reaching the grid spacing, but the grid cell is also of the order of the Debye length, where the vortices cease to be flute-like. The breathing modulates the cat’s eye at a frequency several times its own trapping frequency, an effect that we have not analysed. In view of all this, we do not claim a lifetime for the vortices in physical units, nor that they would decay in a better-resolved or collisional plasma.

Several of these questions are being decided at the time of writing by the restarts of §2.3. If the kinetic loss is Landau damping, it should not depend on the coefficient of the velocity-space hyperdiffusion. If the perpendicular hyperdiffusion erodes the strips, the decay should speed up and slow down with it. If the loss of the instability is set by the grid, it should move to an energy of the vortices about sixteen times lower when the radial resolution is doubled, while a Debye-scale effect would not move. Beyond these, the scaling of the kinetic loss with $k_{\perp}\lambda_D$, of which the production run gives a first indication, and the dependence of (6.1) on the shear, are predictions that further simulations can test. Whether trapping-saturated shear instability, drained by Landau damping at the Debye scale, is what ultimately limits the lifetime of coherent structures in strongly driven pair-plasma condensates — and how the answer changes with physical collisions and with the radial scale of the condensate in a device — appears to be a promising subject for future work.

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Declaration of interests

The authors report no conflict of interest.

Appendix A. Energy integrals of the uniform-flow limit

Here we collect the energy integrals behind (4.10) and (4.11), which were derived in the earlier note and checked independently, as promised in §4.3. For $\hat{\omega}_d > 0$, with $z = \tilde{\omega}/\hat{\omega}_d$ as in §4.3, define

$$\mathcal{K}_n(z) = \int_0^\infty \frac{p(u) u^n}{z - u} du, \quad (\text{A } 1)$$

with the causal continuation from $\text{Im } z > 0$. Then $\mathcal{K}_{n+1} = z\mathcal{K}_n - \langle u^n \rangle$, in particular $\mathcal{K}_1 = z\mathcal{K}_0 - 1$, and, with $u = s^2$,

$$\mathcal{K}_0(z) = \frac{4}{\sqrt{\pi}} \int_0^\infty \frac{s^2 e^{-s^2}}{z - s^2} ds = -2 [1 + \sqrt{z} \mathcal{Z}(\sqrt{z})], \quad \mathcal{Z}(\zeta) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^\infty \frac{e^{-s^2}}{s - \zeta} ds, \quad (\text{A } 2)$$

where $\mathcal{Z}$ is the plasma dispersion function (Fried & Conte 1961), $\mathcal{Z}(\zeta) = i\sqrt{\pi} e^{-\zeta^2} \text{erfc}(-i\zeta)$ for $\text{Im } \zeta > 0$, and $\sqrt{z}$ is the principal root; since $\mathcal{Z}$ is entire, this continues correctly across $\text{Re } z > 0$. On $\text{Re } z < 0$, the non-resonant side, $\mathcal{K}_0$ is real and analytic across the axis.[†] For $\hat{\omega}_d < 0$, the causal prescription becomes $z - i0$, i.e., $\mathcal{K}_0$ is replaced by its complex conjugate; this caveat applies to (A 2), not to the absorptive part below. In the uniform-flow model, the positron and electron contributions to (4.10) are

$$\mathcal{P} = (z - \Omega_* \vartheta) \mathcal{K}_0(z) - \Omega_* \eta \mathcal{K}_1(z), \quad \mathcal{E} = -[(z + \Omega_* \vartheta) \mathcal{K}_0(-z) + \Omega_* \eta \mathcal{K}_1(-z)], \quad (\text{A } 3)$$

where $\vartheta \equiv 1 - 3\eta/2$, and $\mathcal{D} = 1 + \kappa^2 - (\mathcal{P} + \mathcal{E})/2$. For an energy weight $b_0 + b_1 u$ real at real frequency, the absorptive part of the resonant integral is, for either sign of $\hat{\omega}_d$,

$$\text{Im} \int_0^\infty \frac{p(u)(b_0 + b_1 u)}{\omega_r - kU - \hat{\omega}_d u + i0} du = -\frac{2\sqrt{\pi}}{|\hat{\omega}_d|} u_r^{1/2} e^{-u_r} (b_0 + b_1 u_r) \Theta(u_r), \quad (\text{A } 4)$$

$$u_r = \frac{\omega_r - kU}{\hat{\omega}_d},$$

where Θ is the Heaviside function; by (4.4), b_0 must be built from $\omega_r - kU$, not from ω_r . The identity (A 2) agrees with direct quadrature to about 10^{-12} at complex and real z , including $z = x + i0$ and $\text{Re } z < 0$; (A 4) agrees with the $\mathcal{K}$ -function evaluation and with finite- ϵ quadrature for $\hat{\omega}_d = \pm 1.5$; and (4.20) agrees with quadrature to about 10^{-9} .

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[†] In a numerical implementation, at real $z < 0$ with a signed -0 imaginary part, the principal root selects the wrong sheet; the imaginary part should be set to $+0$.

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